Advanced Quantum Mechanics

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Supplementary track, 2013 Fall
Transcript-derived notes curated by LazyingArt LLC

Original lectures by Leonard Susskind. Transcript-derived course notes curated by LazyingArt LLC with Video2Book.

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Chapter 1

Advanced Quantum Mechanics I: Quantum Dynamics and Symmetries

This chapter rebuilds the basic mathematical spine of the lecture: quantum states as vectors, particles in the position representation, the Hamiltonian-generated unitary time flow, and the symmetry principle that identifies conserved quantities by commuting operators. The goal is to move from abstract postulates to the concrete rules used in practical calculations, with a consistent convention and clear links between infinitesimal generators and finite transformations.

1.1 Review of the quantum kinematic structure

A physical system is represented by vectors in a (complex) vector space. A state is written as a ket $|\psi\rangle$, and its dual by $\langle\psi|$. The inner product $\langle\phi|\psi\rangle$ is a complex number associated to each pair of states.

$$\langle\psi|\psi\rangle = 1 \quad \text{for a normalized state.} \tag{1.1}$$

An observable A is represented by a Hermitian operator $\hat{A}$ ($\hat{A}^\dagger = \hat{A}$). The eigenvalue problem has physical meaning:

$$\hat{A}\,|a\rangle = a\,|a\rangle\,, \qquad a \in \mathbb{R}. \tag{1.2}$$

If the system is in $|a\rangle$, measurement of A is certain to give a. Operational distinguishability is encoded by orthogonality:

$$|\phi\rangle \text{ and } |\psi\rangle \text{ are distinguishable with certainty } \iff \langle\phi|\psi\rangle = 0. \tag{1.3}$$

In finite dimensions these are ordinary vectors; for particles and fields, the basis index becomes continuous and states are represented by functions.

1.2 Particle kinematics in the position representation

For a one-dimensional particle, define position eigenstates $|x\rangle$ by

$$\langle x|x'\rangle = \delta(x - x'). \tag{1.4}$$

The corresponding wavefunction is the position amplitude

$$\psi(x) = \langle x|\psi\rangle. \tag{1.5}$$

Its modulus squared is the position probability density:

$$p(x) = |\psi(x)|^2 = \psi^*(x)\psi(x), \tag{1.6}$$

and for normalizable states $\int dx\, |\psi(x)|^2 = 1$.

The position operator acts by multiplication:

$$(\hat{x}\psi)(x) = x\,\psi(x). \tag{1.7}$$

Distributional caveat. The "vectors" $|x\rangle$ are not ordinary square-integrable functions. Their coordinate overlap is the Dirac delta, and the position eigenfunctions are idealized delta spikes rather than physical packet states. Physical states are square-integrable approximants (e.g., narrow Gaussians).

1.3 Momentum in the same representation

The momentum observable is represented by a differential operator:

$$(\hat{p}\psi)(x) = -i\hbar\,\frac{d\psi(x)}{dx}. \tag{1.8}$$

(If $\hbar = 1$, drop it from formulas.)

Momentum eigenstates satisfy

$$\hat{p}\phi_p(x) = p\,\phi_p(x). \tag{1.9}$$

A convenient representative is

$$\phi_p(x) = \frac{1}{\sqrt{2\pi\hbar}}\,e^{ipx/\hbar}, \tag{1.10}$$

which has $|\phi_p(x)|^2 =$ constant: momentum certainty implies complete delocalization in position. This is the uncertainty tradeoff in precise mathematical form.

1.4 Time evolution as a unitary transformation

Time evolution is represented by a family $U(t)$ of operators mapping each state to a later state:

$$|\psi(t)\rangle = U(t)\,|\psi(0)\rangle\,, \qquad U(0) = \mathbf{1}. \tag{1.11}$$

A symmetry of information requires inner products to be preserved:

$$\langle\phi|\psi\rangle = \langle U(t)\phi|U(t)\psi\rangle \implies U^\dagger(t)U(t) = \mathbf{1}. \tag{1.12}$$

So $U(t)$ is unitary.

Assuming continuity, for small ϵ,

$$U(\epsilon) = \mathbf{1} - i\epsilon H + \mathcal{O}(\epsilon^2). \tag{1.13}$$

Unitarity to first order gives

$$H^\dagger = H, \tag{1.14}$$

so H is Hermitian and identified as the Hamiltonian.

Now use

$$|\psi(t+\epsilon)\rangle = U(\epsilon)\,|\psi(t)\rangle \tag{1.15}$$

and rearrange:

$$\frac{|\psi(t+\epsilon)\rangle - |\psi(t)\rangle}{\epsilon} = -iH\,|\psi(t)\rangle + \mathcal{O}(\epsilon). \tag{1.16}$$

Hence

$$i\,\frac{d}{dt}\,|\psi(t)\rangle = H\,|\psi(t)\rangle\,. \tag{1.17}$$

For wavefunctions this is

$$i\hbar\,\frac{\partial\psi(x,t)}{\partial t} = H\psi(x,t). \tag{1.18}$$

Finite time is obtained by exponentiation:

$$U(t) = e^{-iHt/\hbar}. \tag{1.19}$$

Stationary states are eigenstates of H:

$$H\,|E_n\rangle = E_n\,|E_n\rangle\,. \tag{1.20}$$

Then

$$|E_n(t)\rangle = e^{-iE_n t/\hbar}\,|E_n\rangle\,, \tag{1.21}$$

so the same Hamiltonian controls both time dependence and allowed energy levels.

1.5 Symmetry as commuting operators

A symmetry V is taken to be a unitary map on states that preserves all physical distinctions. If it represents a true symmetry of dynamics, then evolving then transforming is equivalent to transforming then evolving:

$$V\,U(t) = U(t)\,V, \qquad \forall t. \tag{1.22}$$

Using $U(t) = e^{-iHt/\hbar}$, this is equivalent to

$$[V, H] = 0. \tag{1.23}$$

For a continuous family $V(\epsilon) = \mathbf{1} - i\epsilon G + \mathcal{O}(\epsilon^2)$, the same logic gives

$$[G, H] = 0. \tag{1.24}$$

When $G = G^\dagger$, this is exactly a conserved generator:

$$\frac{d}{dt}\langle G\rangle = \frac{i}{\hbar}\langle [H, G]\rangle = 0. \tag{1.25}$$

Thus symmetry $\leftrightarrow$ conserved quantity is the operator-level statement.

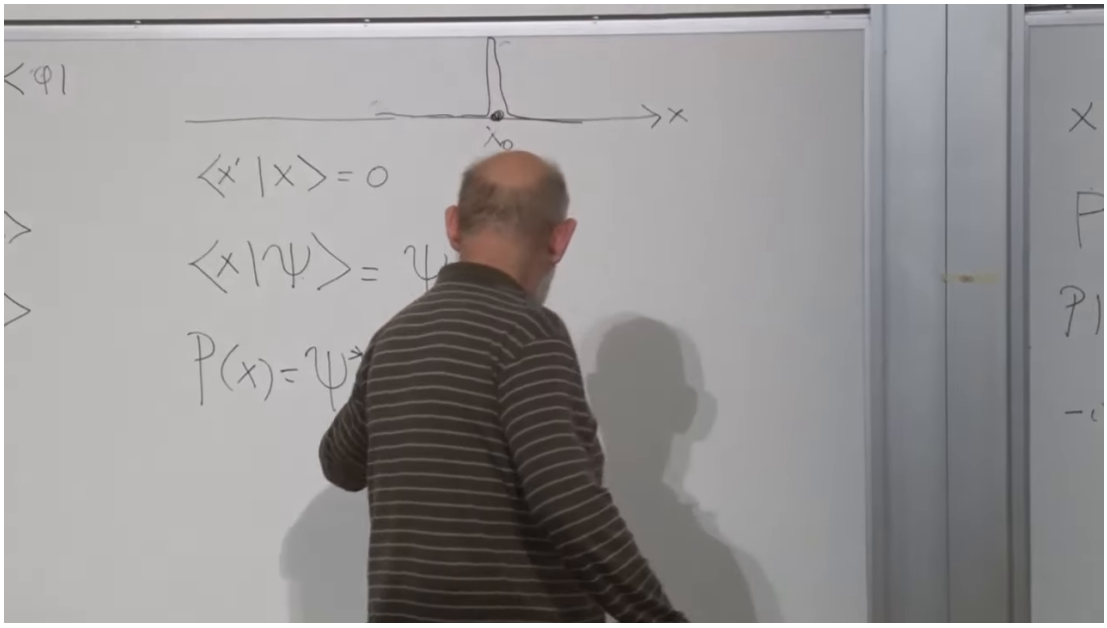


Figure 1.1: Infinitesimal-to-finite evolution structure for $U(\epsilon)$.

$$
\begin{tikzcd}
|\psi_1\rangle \arrow[r, "U(t)"] \arrow[d, "V"'] & |\psi_2\rangle = U(t)\,|\psi_1\rangle \arrow[d, "V"] \\
V\,|\psi_1\rangle \arrow[r, "U(t)"] & V\,|\psi_2\rangle = U(t)\,V\,|\psi_1\rangle
\end{tikzcd}
$$

Figure 1.2: A symmetry operation V commutes with time evolution $U(t)$ if the two routes between states coincide.

1.6 Continuous symmetry example: spatial translations

Define the spatial translation operator $T(\epsilon)$ acting on wavefunctions by

$$(T(\epsilon)\psi)(x) = \psi(x+\epsilon), \tag{1.26}$$

which is a rightward shift by ϵ. For small ϵ,

$$\psi(x+\epsilon) = \psi(x) + \epsilon\,\partial_x\psi(x) + \mathcal{O}(\epsilon^2), \tag{1.27}$$

so

$$T(\epsilon) = \mathbf{1} + \epsilon\,\partial_x + \mathcal{O}(\epsilon^2) = \mathbf{1} - \frac{i\epsilon}{\hbar}\,\hat{p}_x + \mathcal{O}(\epsilon^2). \tag{1.28}$$

Hence

$$T(\epsilon) = \exp\left(-\frac{i}{\hbar}\epsilon\,\hat{p}_x\right), \tag{1.29}$$

so the generator is

$$G_{\text{trans}} = \frac{\hat{p}_x}{\hbar}. \tag{1.30}$$

For a free particle,

$$H = \frac{\hat{p}_x^2}{2m}, \tag{1.31}$$

and $[H, \hat{p}_x] = 0$, so translation is a symmetry and $\hat{p}_x$ is conserved. Thus the same operator appears as both "generator of translations" and "conserved quantity," illustrating the lecture's central symmetry–conservation link.

1.7 Summary

The lecture's core chain is: state vectors → continuous basis amplitudes → operator representation → unitary time flow. Continuity plus unitarity forces $U(t)$ to be generated by a Hermitian Hamiltonian and yields the Schrödinger equations. A symmetry is an additional unitary map compatible with $U(t)$, which becomes a commutator condition with H. For continuous symmetries, infinitesimal generators G define finite transformations via exponentials, and commuting generators are conserved quantities. Translations provide the prototype: $T(\epsilon) = e^{-i\epsilon\hat{p}_x/\hbar}$, and for $H = p^2/2m$, momentum conservation follows directly from $[H, \hat{p}_x] = 0$.

Chapter 2

Symmetry, Degeneracy, and Angular Momentum for Quantum Rotational Systems

This chapter develops a symmetry-first view of rotational quantum systems. We define symmetries as operators commuting with the Hamiltonian, distinguish generic level coincidences from symmetry-protected multiplets, and build angular-momentum structure from the commutator algebra of generators. The particle on a circle provides an explicit worked model where quantized m-values, reflection pairing, and ladder operators emerge from operator algebra rather than ad hoc assumptions.

2.1 Symmetry as an invariance principle

A symmetry is an operation that leaves the quantum description unchanged. For a unitary transformation S, invariance of dynamics is the condition

$$SHS^\dagger = H$$

equivalently

$$[H, S] = 0.$$

Because S commutes with H, Schrödinger evolution is covariant under S: if $|\psi(t)\rangle$ evolves by H, then $S|\psi(t)\rangle$ evolves in the same way. In particular, E-eigenvalues are mapped to the same E-eigenspaces.

Symmetries can be exact (invariance of the full Hamiltonian), approximate (effective in a regime), or accidental (valid only for very specific parameter choices). Our focus is on exact symmetries because they control spectral structure.

2.2 Symmetry examples and the organizing role of operators

The same formal rule $[H, S] = 0$ unifies many familiar examples:

- spatial translations and rotations,

- interchange of identical particles,
- relabeling symmetries of a periodic lattice (discrete translation in crystal index),
- discrete mirror or inversion operations.

In practice, this is less about geometry alone and more about algebraic leverage: a set of commuting operators gives you a good quantum-number basis, while noncommuting families often generate nontrivial multiplets. Symmetry language gives a compact way to classify allowed degeneracies before solving differential equations.

Figure 2.1: Prototype symmetry motifs: geometric and discrete transformations that can commute with the Hamiltonian.

2.3 Degeneracy, symmetry, and when degeneracy is protected

For a bounded discrete spectrum, exact equality of two nearby energies is rare unless there is structure behind it. We call this *accidental* coincidence when no symmetry explanation exists. In contrast, symmetry-protected degeneracies are stable under small perturbations that preserve the same symmetry.

Proposition 2.1. *Let A and B be Hermitian symmetry generators commuting with H: $[A, H] = [B, H] = 0$. Define*

$$C := \frac{1}{i\hbar}[A, B].$$

Then C is Hermitian and $[C, H] = 0$.

Proof. Since A and B are Hermitian, $[A, B]^\dagger = -[A, B]$, so C is Hermitian. Then

$$[C, H] = \frac{1}{i\hbar}([AB, H] - [BA, H])$$

and expanding each commutator gives zero because both A and B commute with H. Hence $[C, H] = 0$. $\square$

Thus one can grow a closed set of symmetry generators by commutators. If that set closes to a Lie algebra with nontrivial structure constants, the representation theory is what drives degeneracy and spectral multiplets.

2.4 Particle on a circle: quantized L_z from single-valued wave functions

Let $\theta \in [0, 2\pi)$ label the position of a particle constrained to a circle. A small rotation by ϵ acts as

$$(\mathcal{R}(\epsilon)\psi)(\theta) = \psi(\theta - \epsilon).$$

Its infinitesimal form defines the generator L_z:

$$\mathcal{R}(\epsilon)\psi(\theta) \approx \left(1 - \frac{i\epsilon}{\hbar} L_z\right)\psi(\theta), \qquad L_z = -i\hbar\frac{d}{d\theta}.$$

Hence

$$L_z \left|\psi_m\right\rangle = m\hbar \left|\psi_m\right\rangle \quad \Longleftrightarrow \quad -i\hbar\frac{d\psi_m}{d\theta} = m\hbar\psi_m$$

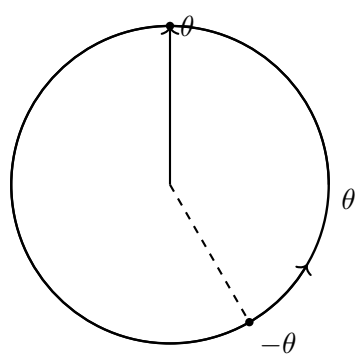
and therefore

$$\psi_m(\theta) = \frac{1}{\sqrt{2\pi}} e^{im\theta}.$$

Single-valuedness on $0 \sim 2\pi$ gives

$$\psi_m(\theta + 2\pi) = \psi_m(\theta) \iff e^{i2\pi m} = 1 \iff m \in \mathbb{Z}.$$

So the orbital angular-momentum quantum numbers are $m = 0, \pm 1, \pm 2, \ldots$.



Identification of θ and $\theta + 2\pi$: $\psi(\theta + 2\pi) = \psi(\theta)$

Figure 2.2: Circle coordinate, opposite angles, and periodicity condition.

2.5 Mirror symmetry and the $m \leftrightarrow -m$ pairing

Introduce a mirror (reflection) operator $\mathcal{M}$ with

$$(\mathcal{M}\psi)(\theta) = \psi(-\theta).$$

Its action on a momentum eigenmode is

$$\mathcal{M}\psi_m(\theta) = \psi_{-m}(\theta).$$

On a circle eigenstate:

$$\mathcal{M}L_z\psi_m = m\hbar\,\mathcal{M}\psi_m, \qquad L_z\mathcal{M}\psi_m = -m\hbar\,\mathcal{M}\psi_m,$$

so

$$[\mathcal{M}, L_z]\psi_m = 2m\hbar\,\mathcal{M}\psi_m \neq 0 \quad (m \neq 0).$$

Thus $\mathcal{M}$ does not commute with L_z, even though both may commute with H in a symmetric setup. When both symmetries are present (no mirror-breaking field), states m and $-m$ are paired at equal energy.

Important warning from the formalism: equal probabilities $|\psi_m(\theta)|^2 = |\psi_{-m}(\theta)|^2$ does not imply identical wave functions. Degeneracy concerns states and operators, not only densities.

A uniform magnetic field is a standard counterexample: it can split $\pm m$ levels while preserving some rotational structure of orbital motion in other settings. Symmetry arguments apply to what the Hamiltonian does, not to geometric intuition alone.

2.6 Generators, commutator algebra, and closure

For continuous symmetries a small transformation is written as

$$U(\epsilon) = 1 - \frac{i\epsilon}{\hbar}G + \mathcal{O}(\epsilon^2),$$

with Hermitian generator G. The commutator of two generators is another generator of another direction in the same symmetry structure if closure holds.

A useful worked pattern is:

$$[\,\mathcal{M}, L_z\,] \neq 0, \qquad \text{while } \; [H, L_z] = 0, \; [H, \mathcal{M}] = 0 \text{ (if mirror symmetric)}.$$

Hence $\mathcal{M}$ and L_z together generate two operations whose noncommutativity forces additional structure.

2.7 Three-dimensional angular momentum and ladder structure

For a particle in 3D,

$$L_x = yp_z - zp_y, \qquad L_y = zp_x - xp_z, \qquad L_z = xp_y - yp_x,$$

with canonical $[x_i, p_j] = i\hbar\delta_{ij}$. A compact expansion gives

$$[L_x, L_y] = i\hbar L_z,$$

and cyclic permutations. Equivalently,

$$[L_i, L_j] = i\hbar \sum_k \epsilon_{ijk} L_k.$$

For a rotationally invariant Hamiltonian, $[H, L_i] = 0$, so L_x, L_y, L_z are conserved quantities.

Define raising and lowering operators

$$L_\pm := L_x \pm iL_y.$$

Using the algebra above,

$$[L_z, L_\pm] = \pm\hbar L_\pm, \qquad [L_+, L_-] = 2\hbar L_z.$$

If $|\ell, m\rangle$ is an eigenvector of L_z,

$$L_z|\ell, m\rangle = m\hbar|\ell, m\rangle,$$

then

$$L_z(L_\pm|\ell, m\rangle) = (m \pm 1)\hbar(L_\pm|\ell, m\rangle),$$

so $L_\pm$ create a state with adjacent m-value when nonzero.

In a finite ladder representation, the chain terminates:

$$L_+|\ell, \ell\rangle = 0, \qquad L_-|\ell, -\ell\rangle = 0,$$

giving $m = -\ell, -\ell+1, \ldots, \ell$, hence $2\ell+1$ states in one rotational multiplet. For orbital wave functions on configuration space, $\ell, m \in \mathbb{Z}$ and $m \in \{-\ell, \ldots, \ell\}$. For more general angular-momentum types (including spin), half-integer m spectra are also possible; in this lecture the orbital circle argument selects the integer branch.

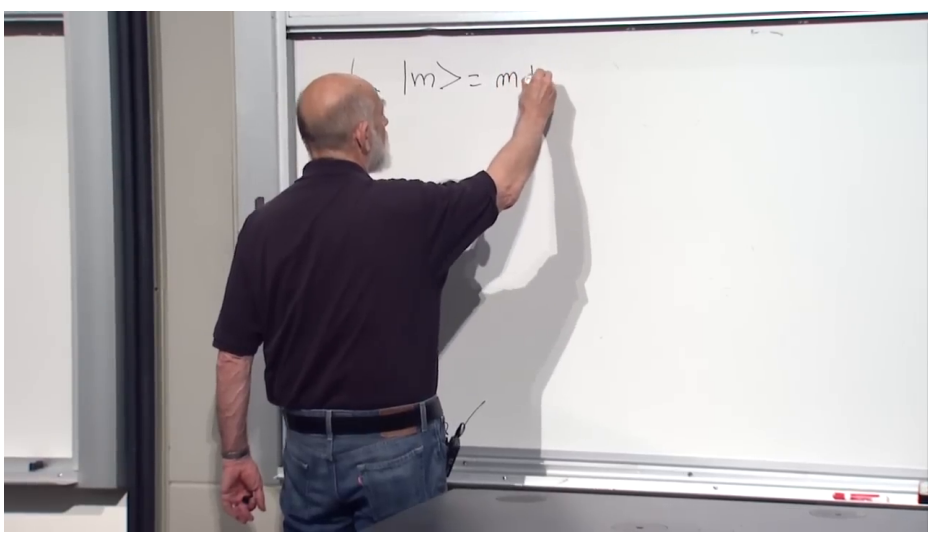


Figure 2.3: Illustrative ladder picture: $L_\pm$ connect adjacent m-levels inside a symmetry multiplet.

2.8 Summary

Symmetry enters as a commutation criterion with the Hamiltonian. Single symmetries can explain invariant spectral structure, but genuine multiplets require a noncommuting symmetry set whose commutator algebra closes. The circle provides the cleanest demonstration: rotational invariance gives $L_z = -i\hbar\partial_\theta$, single-valuedness gives integer m, and mirror symmetry links m to $-m$. In three dimensions, the algebra $[L_i, L_j] = i\hbar\epsilon_{ijk}L_k$ yields ladder operators $L_\pm$, a finite m-tower for each ℓ, and equal-energy multiplets in rotationally invariant systems.

Chapter 3

Angular Momentum, Central Forces, and the Oscillator Ladder

These notes follow Leonard Susskind's *Advanced Quantum Mechanics* Lecture 3, curated by LazyingArt LLC. The lecture opens as a continuation of the previous discussion of angular momentum, but almost immediately shifts from algebra to physics: angular momentum is the language of how a wavefunction depends on angles. From there the hour unfolds in two acts. First, angular-momentum multiplets are reconstructed and then used to reduce the central-force Schrödinger equation to a one-dimensional radial problem. Second, the same ladder-operator instinct reappears in the harmonic oscillator. Throughout, Susskind suppresses factors of $\hbar$ on the board; they are restored here.

3.1 Angular Momentum as Angular Dependence of the Wavefunction

Susskind begins by recalling the angular-momentum operators and the multiplets they form, but he does not stay in recap mode for long. Instead he asks what angular momentum is really about. His example is a particle moving in a central force field. Classically, the angular momentum

$$\mathbf{L} = \mathbf{r} \times \mathbf{p} \tag{3.1}$$

is conserved, so both its magnitude and its direction remain fixed. The orbit therefore stays in a fixed plane perpendicular to $\mathbf{L}$.

Quantum mechanically the particle is described by a wavefunction, and the point is no longer the orbit itself but the way the wavefunction changes under rotation. In a central-force problem the wavefunction may be written as a function of radius and angles,

$$\Psi = \Psi(r, \theta, \phi), \tag{3.2}$$

and angular momentum acts only on the angular part.

A two-dimensional prototype. Susskind briefly returns to two dimensions because there the basic idea is completely transparent. If motion is confined to a plane, the only relevant angular

momentum is the generator of rotations about the perpendicular axis:

$$L_z = -i\hbar \frac{\partial}{\partial \theta}. \tag{3.3}$$

An angular-momentum eigenstate therefore satisfies

$$L_z \psi(r,\theta) = \hbar \ell \, \psi(r,\theta), \tag{3.4}$$

or

$$-i\hbar \frac{\partial}{\partial \theta} \psi(r,\theta) = \hbar \ell \, \psi(r,\theta). \tag{3.5}$$

Here the derivative is only with respect to θ. The radial coordinate is a spectator. Solving for the θ-dependence gives

$$\psi(r,\theta) = \chi(r) e^{i\ell\theta}, \tag{3.6}$$

where ℓ must be an integer so that the wavefunction returns to itself when θ goes around the circle by 2π.

This is the first structural lesson of the lecture. Angular momentum is determined by the angular dependence of the wavefunction. The radial factor $\chi(r)$ may matter for the energy, but it does not matter for the angular-momentum eigenvalue.

Back to three dimensions. The same statement survives in three dimensions. One writes the wavefunction schematically as

$$\Psi(r,\theta,\phi) = R(r)\, Y_{\ell m}(\theta,\phi), \tag{3.7}$$

where the angular factor $Y_{\ell m}$ carries the rotational quantum numbers. These are the spherical harmonics. Susskind's emphasis is that one need not know their explicit formulas at this stage. What matters is that the angular problem and the radial problem separate, and angular momentum belongs entirely to the first of them.

With that physical point fixed, he returns to the operator algebra.

3.2 From the L_z Basis to Finite Multiplets

The next move, as Susskind says in a distinctly Dirac-like spirit, is to pick one component of angular momentum and work in its eigenbasis. Take L_z. The basic commutators are

$$[L_x, L_y] = i\hbar L_z, \qquad [L_y, L_z] = i\hbar L_x, \qquad [L_z, L_x] = i\hbar L_y. \tag{3.8}$$

Now define

$$L_\pm = L_x \pm i L_y. \tag{3.9}$$

A short computation gives

$$[L_z, L_\pm] = \pm\hbar L_\pm. \tag{3.10}$$

Therefore, if

$$L_z|\ell, m\rangle = \hbar m \, |\ell, m\rangle, \tag{3.11}$$

then

$$L_z\big(L_\pm|\ell,m\rangle\big) = \big(L_\pm L_z + [L_z, L_\pm]\big)|\ell,m\rangle \tag{3.12}$$

$$= \hbar(m \pm 1)\big(L_\pm|\ell,m\rangle\big). \tag{3.13}$$

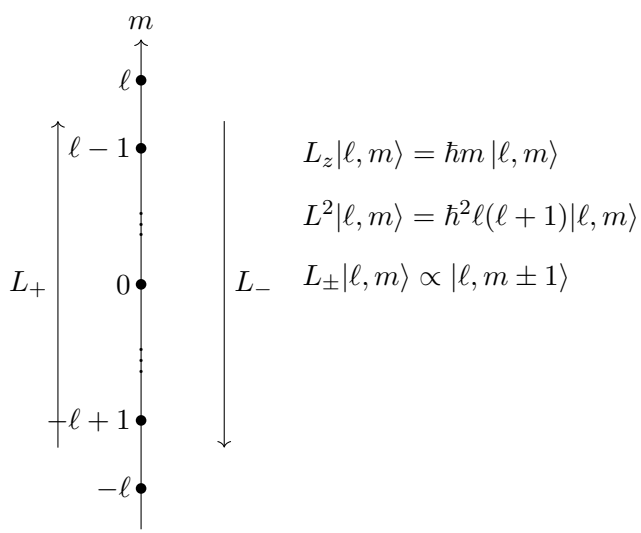


Figure 3.1: The finite angular-momentum ladder suggested by the blackboard discussion. The ladder operators move in unit steps of m, and the multiplet terminates symmetrically at $m = \pm\ell$.

So L_+ raises m by one unit and L_- lowers it by one unit:

$$L_\pm|\ell, m\rangle \propto |\ell, m \pm 1\rangle. \tag{3.14}$$

That is the first algebraic payoff. The second is the question Susskind immediately asks: does the ladder continue forever? It can only stop if eventually one of the raised or lowered states vanishes:

$$L_+|\text{top}\rangle = 0, \qquad L_-|\text{bottom}\rangle = 0. \tag{3.15}$$

Rotational symmetry then forces the spectrum to be symmetric about zero. There is nothing fundamentally privileged about the $+z$-direction relative to the $-z$-direction. Hence a finite ladder must run from some highest value $m = \ell$ down to the reflected value $m = -\ell$:

$$m = -\ell, -\ell + 1, \ldots, \ell - 1, \ell. \tag{3.16}$$

The number of states is therefore

$$2\ell + 1. \tag{3.17}$$

Susskind makes one further observation here. If the steps are always in integers, then a finite symmetric ladder can begin either at an integer or at a half-integer. Thus the abstract angular-momentum algebra allows both integer and half-integer multiplets. Later, when the discussion specializes to orbital wavefunctions on the sphere, only integer ℓ remain.

Remark 3.1. The lecture briefly mentions the possibility of infinite ladders. They are mathematically conceivable, but they are set aside as physically pathological for the finite-energy situations of interest here.

3.3 What a Multiplet Has in Common

Once the ladder has been built, Susskind asks what characterizes the entire multiplet. The answer is the square of the total angular momentum,

$$L^2 = L_x^2 + L_y^2 + L_z^2. \tag{3.18}$$

To evaluate it, he rewrites $L_x^2 + L_y^2$ in terms of the ladder operators. The ordered product is

$$\begin{aligned} L_- L_+ &= (L_x - iL_y)(L_x + iL_y) && (3.19)\\ &= L_x^2 + L_y^2 + iL_xL_y - iL_yL_x && (3.20)\\ &= L_x^2 + L_y^2 + i[L_x, L_y] && (3.21)\\ &= L_x^2 + L_y^2 - \hbar L_z. && (3.22) \end{aligned}$$

Therefore

$$L^2 = L_z^2 + \hbar L_z + L_- L_+. \tag{3.23}$$

There is also the equally valid alternative

$$L^2 = L_z^2 - \hbar L_z + L_+ L_-, \tag{3.24}$$

which corresponds to factoring in the opposite order. An audience member asks precisely this question in the lecture, and Susskind's answer is that the same derivation works if one applies the reordered identity to the bottom state instead of the top.

Worked derivation of $L^2 = \hbar^2 \ell(\ell+1)$. Apply the first ordered form to the highest state of the multiplet:

$$\begin{aligned} L^2|\ell,\ell\rangle &= \left(L_z^2 + \hbar L_z + L_- L_+\right)|\ell,\ell\rangle && (3.25)\\ &= \left(\hbar^2\ell^2 + \hbar^2\ell\right)|\ell,\ell\rangle + L_-(L_+|\ell,\ell\rangle). && (3.26) \end{aligned}$$

By definition of the top state,

$$L_+|\ell,\ell\rangle = 0, \tag{3.27}$$

and so

$$L^2|\ell,\ell\rangle = \hbar^2\ell(\ell+1)|\ell,\ell\rangle. \tag{3.28}$$

This is the blackboard formula $L^2 = \ell(\ell+1)$ with the suppressed factor of $\hbar^2$ restored.

The lecture then pushes one step further. This is not just a property of the top state. In fact,

$$[L^2, L_i] = 0, \qquad i = x, y, z, \tag{3.29}$$

and therefore L^2 commutes with $L_\pm$ as well. For example,

$$\begin{aligned} L^2(L_-|\ell,\ell\rangle) &= L_-(L^2|\ell,\ell\rangle) && (3.30)\\ &= \hbar^2\ell(\ell+1)(L_-|\ell,\ell\rangle). && (3.31) \end{aligned}$$

So the next state down has the same L^2-eigenvalue, and the same argument propagates all the way to the bottom of the ladder.

This is the meaning of the multiplet. All $2\ell+1$ states have the same total angular momentum, but different projections on the z-axis:

$$L^2|\ell,m\rangle = \hbar^2\ell(\ell+1)|\ell,m\rangle, \qquad L_z|\ell,m\rangle = \hbar m\,|\ell,m\rangle. \tag{3.32}$$

In the orbital case these states are represented by the spherical harmonics $Y_{\ell m}(\theta,\phi)$, with $\ell = 0, 1, 2, \ldots$ and $m = -\ell, \ldots, \ell$.

Susskind then makes the transition that matters for atomic physics. In a central-force problem the Hamiltonian commutes with angular momentum, so all members of a fixed ℓ-multiplet have the same energy:

$$[H, L_i] = 0 \qquad \text{for } V = V(r). \tag{3.33}$$

That closes the angular-momentum recap and opens the central-force problem.

3.4 From Three Dimensions to an Effective Radial Problem

At this point Susskind says explicitly that he will use a little classical intuition, even a tiny bit of cheating, to motivate the reduction of the central-force problem. The starting Hamiltonian is

$$H = \frac{p^2}{2m} + V(r), \tag{3.34}$$

with p^2 the full three-dimensional momentum squared and $V(r)$ depending only on the distance from the origin.

Classical reduction. Because the force is central, angular momentum is conserved. Classically one may therefore choose coordinates so that the orbit lies in the xy-plane. In that plane decompose the momentum into a radial component p_r and an angular component p_θ:

$$p^2 = p_r^2 + p_\theta^2. \tag{3.35}$$

The Hamiltonian becomes

$$H = \frac{p_r^2 + p_\theta^2}{2m} + V(r). \tag{3.36}$$

Now use the magnitude of the classical angular momentum:

$$L = r\, p_\theta. \tag{3.37}$$

Hence

$$L^2 = r^2 p_\theta^2, \qquad p_\theta^2 = \frac{L^2}{r^2}. \tag{3.38}$$

Substituting back gives the board equation

$$H = \frac{p_r^2}{2m} + \frac{L^2}{2mr^2} + V(r). \tag{3.39}$$

This is the effective radial Hamiltonian. Once the angular momentum is fixed, everything angular has been frozen into the parameter L, and the remaining motion is one-dimensional in r.

It is natural to define the effective potential

$$V_{\text{eff}}(r) = \frac{L^2}{2mr^2} + V(r). \tag{3.40}$$

The extra $1/r^2$ term is the centrifugal barrier. It is repulsive, not attractive: as r decreases, the potential rises sharply, so the corresponding force pushes the particle away from the origin.

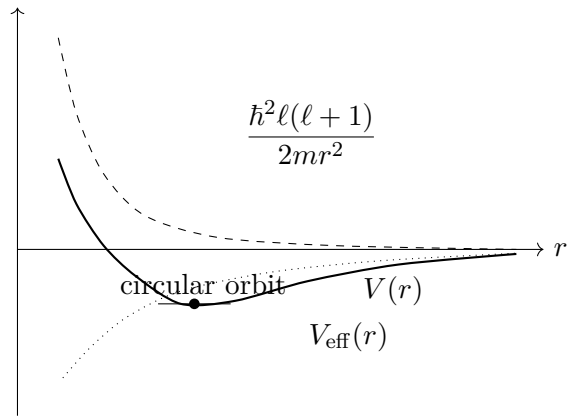


Figure 3.2: The effective radial potential for an attractive central force. The Coulomb tail $V(r) \sim -1/r$ and the centrifugal barrier $\sim 1/r^2$ combine into a one-dimensional potential with a minimum when $\ell \neq 0$.

What the effective potential means. For a Coulomb attraction $V(r) \sim -1/r$, the barrier behaves as $+1/r^2$. Near the origin the $1/r^2$ term dominates; far away the $1/r$ term dominates. Their sum has a minimum. A particle sitting at that minimum executes a circular orbit, while a particle with slightly different energy oscillates radially between a minimum and maximum radius. In the Coulomb case those radial oscillations are the classical elliptical orbits.

There is one exceptional case, emphasized later in the discussion: if $\ell = 0$, then the centrifugal barrier vanishes. Classically nothing prevents the motion from heading straight through the origin.

3.5 The Quantum Radial Equation

Susskind now turns to the quantum problem, but he keeps the classical reduction in view. He does not carry out every angular derivative on the board. Instead he uses the classical form of the Hamiltonian as a guide and then states the separated quantum equation. The omitted manipulations are real, but they do not change the basic story.

The full time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2}{\partial x^2}+\frac{\partial^2}{\partial y^2}+\frac{\partial^2}{\partial z^2}\right)\Psi + V(r)\Psi = E\Psi. \tag{3.41}$$

Now choose a state of definite angular momentum:

$$\Psi_{\ell m}(r,\theta,\phi) = \psi(r)\, Y_{\ell m}(\theta,\phi). \tag{3.42}$$

Once ℓ and m are fixed, the angular dependence has already been solved by the spherical harmonic. The remaining unknown is the radial function $\psi(r)$.

The radial momentum operator is

$$p_r = -i\hbar\,\frac{d}{dr}, \tag{3.43}$$

so that

$$p_r^2 \to -\hbar^2\,\frac{d^2}{dr^2}. \tag{3.44}$$

Susskind is slightly casual about the sign when he speaks it aloud, but only p_r^2 appears in the Hamiltonian, so the final radial kinetic term is unambiguous.

On a state of definite ℓ, one replaces L^2 by $\hbar^2\ell(\ell+1)$. Thus the radial Hamiltonian acts as

$$H\psi(r) = E\psi(r), \tag{3.45}$$

with

$$H = -\frac{\hbar^2}{2m}\frac{d^2}{dr^2} + \frac{\hbar^2\ell(\ell+1)}{2mr^2} + V(r), \tag{3.46}$$

and therefore

$$-\frac{\hbar^2}{2m}\frac{d^2}{dr^2}\psi(r) + \frac{\hbar^2\ell(\ell+1)}{2mr^2}\psi(r) + V(r)\psi(r) = E\psi(r). \tag{3.47}$$

Remark 3.2. The lecture writes this as a pure second-order equation for $\psi(r)$. In standard separation of variables this clean one-dimensional form is the equation for the reduced radial function $u_\ell(r) = rR_\ell(r)$. To stay close to Susskind's notation while remaining precise, it is best to read $\psi(r)$ here as that reduced radial variable.

This is the main payoff of the angular-momentum recap. The original three-dimensional partial differential equation has become an ordinary differential equation in one variable. The angular structure has not disappeared; it has been compressed into the single parameter ℓ.

There is also a conceptual point that surfaces in the question period. Quantum mechanically one cannot literally rotate the whole wavefunction into a fixed plane, as one does classically for a single orbit. The memory of full three-dimensional motion survives in the fact that the relevant operator is L^2, not L_z^2. If one could replace L^2 by L_z^2, one would get something like ℓ^2. Instead one gets the genuinely three-dimensional result $\ell(\ell+1)$.

Susskind stops short of solving the radial equation explicitly. That, he says, is a matter of differential equations, and the lecture is after the structure of the spectrum rather than the explicit special functions.

3.6 Nodes, Degeneracies, and the Coulomb Accident

Having reduced the problem to one dimension, Susskind turns immediately to what one can say about the spectrum without solving the equation in detail.

Node counting. For fixed ℓ, the radial Schrödinger equation is a one-dimensional bound-state problem. The lowest bound state has no nodes, the first excited state has one node, the next has two, and so on. Here a node means a zero of the radial wavefunction. Susskind gives the physical reason in his usual way: more nodes mean more rapid wiggles; more rapid wiggles mean larger momentum; larger momentum means higher kinetic energy. In this sense the node count organizes the bound-state spectrum.

The generic central-force spectrum. Each value of ℓ gives a different radial equation because the centrifugal term depends on ℓ. Thus one obtains a separate tower of radial states for each ℓ. The $\ell = 0$ tower sits lowest because the centrifugal barrier is absent there. The $\ell = 1$ tower is shifted upward, the $\ell = 2$ tower upward again, and so on. At every level in the ℓ-th tower there are

$$2\ell + 1 \tag{3.48}$$

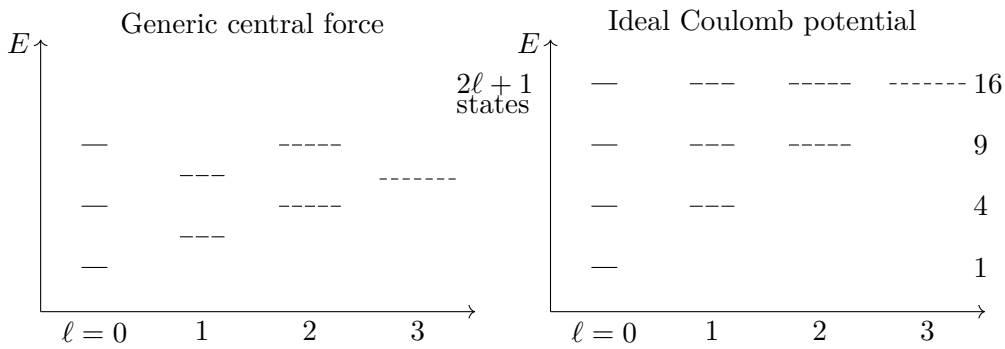


Figure 3.3: Left: the generic qualitative spectrum of an attractive central force. Each value of ℓ has its own radial tower, shifted upward by the centrifugal term, and each level carries degeneracy $2\ell + 1$. Right: in the ideal Coulomb problem, levels from different ℓ-towers align into shells with orbital degeneracies $1, 4, 9, 16, \ldots$.

degenerate states, corresponding to the allowed values of m.

At sufficiently high energy the states cease to be bound. Classically the particle escapes; quantum mechanically the discrete spectrum gives way to the continuum. For the bound states of an attractive potential, the energies lie below that escape threshold.

The special case $\ell = 0$. One audience question returns to the missing centrifugal barrier when $\ell = 0$. Classically, if the potential is Coulombic, the particle would simply run toward the center. Quantum mechanically that does not happen. The lowest state cannot be both sharply localized at the origin and at rest there. The uncertainty principle forces a compromise: the ground-state wavefunction spreads over a finite region, and the corresponding energy is finite rather than negatively infinite.

The Coulomb accident. For the special potential $V(r) \propto -1/r$, Susskind describes a remarkable extra degeneracy. The lowest $\ell = 1$ state has exactly the same energy as the first excited $\ell = 0$ state. The lowest $\ell = 2$ state has the same energy as the second excited $\ell = 0$ state. More generally, the lowest state in the ℓ-tower aligns with the ℓ-node state in the $\ell = 0$ tower. The result is a sequence of shells whose orbital degeneracies are

$$1, \qquad 4, \qquad 9, \qquad 16, \qquad \ldots \tag{3.49}$$

or, equivalently,

$$\sum_{k=0}^{n-1} (2k + 1) = n^2. \tag{3.50}$$

Susskind repeatedly warns that this is special to the ideal nonrelativistic Coulomb problem. A finite nuclear size, relativistic corrections, and spin all spoil it. The hydrogen atom approximately displays the pattern, but not exactly. In particular, because spin has not yet been included, the analysis is missing a factor of two. Real atomic shell counts begin

$$2, \qquad 8, \qquad 18, \qquad 32, \qquad \ldots \tag{3.51}$$

rather than

$$1, \qquad 4, \qquad 9, \qquad 16, \qquad \ldots \tag{3.52}$$

The lecture also pauses on a subtle question about wavefunctions. Bound states do not have compact support. They do not literally vanish beyond some finite radius. Instead they fall off rapidly, typically exponentially, so that the probability of finding the particle very far away is nonzero but extremely small.

That completes the atomic part of the lecture. Susskind then puts atoms aside for the moment and turns to a new system in which the same algebraic luck appears again.

3.7 The Harmonic Oscillator: Dirac's Trick Again

The transition to the harmonic oscillator is not a digression so much as a replay. Susskind emphasizes that many small oscillations in physics are approximately harmonic, and he even notes that oscillations near the bottom of an effective radial well are an example. The oscillator is the second case in which ladder operators solve the spectrum almost at once.

Take the mass to be 1 and write the spring constant as $k = \omega^2$. Then the classical Hamiltonian is

$$H = \frac{p^2}{2} + \frac{\omega^2 x^2}{2}. \tag{3.53}$$

Hamilton's equations are

$$\dot{x} = \frac{\partial H}{\partial p} = p, \qquad \dot{p} = -\frac{\partial H}{\partial x} = -\omega^2 x. \tag{3.54}$$

Therefore

$$\ddot{x} = -\omega^2 x, \tag{3.55}$$

which identifies ω as the classical frequency of oscillation.

Factorization and the zero-point term. Now comes the Dirac-like move. Since the Hamiltonian is a sum of squares, try to factor it:

$$\frac{1}{2}(p + i\omega x)(p - i\omega x). \tag{3.56}$$

Quantum mechanically this is not exactly H, because x and p do not commute. Using

$$[x, p] = i\hbar, \tag{3.57}$$

one finds

$$\frac{1}{2}(p + i\omega x)(p - i\omega x) = \frac{1}{2}\big(p^2 + \omega^2 x^2 + i\omega[x, p]\big) \tag{3.58}$$

$$= \frac{1}{2}\big(p^2 + \omega^2 x^2 - \hbar\omega\big). \tag{3.59}$$

Hence

$$H = \frac{1}{2}(p + i\omega x)(p - i\omega x) + \frac{1}{2}\hbar\omega. \tag{3.60}$$

The additive constant $\frac{1}{2}\hbar\omega$ is the zero-point energy. In the lecture Susskind temporarily drops it in order to focus on the ladder structure, and then restores it at the end.

Define

$$A_+ = \frac{p + i\omega x}{\sqrt{2\hbar\omega}}, \qquad A_- = \frac{p - i\omega x}{\sqrt{2\hbar\omega}}, \tag{3.61}$$

so that A_+ and A_- are Hermitian conjugates of one another. In standard notation they are $a^\dagger$ and a. The Hamiltonian becomes

$$H = \hbar\omega\left(A_+A_- + \frac{1}{2}\right). \tag{3.62}$$

Now define the number operator

$$N = A_+A_-. \tag{3.63}$$

At this point Susskind asks what Dirac would do. His answer is immediate: commute the factors. A direct calculation gives

$$[A_-, A_+] = 1. \tag{3.64}$$

From this it follows that

$$[N, A_+] = A_+, \qquad [N, A_-] = -A_-. \tag{3.65}$$

The ladder. Suppose

$$N|n\rangle = n|n\rangle. \tag{3.66}$$

Then

$$N(A_+|n\rangle) = (A_+N + [N, A_+])|n\rangle \tag{3.67}$$
$$= (n+1)(A_+|n\rangle), \tag{3.68}$$

so A_+ raises the eigenvalue by one. Similarly,

$$N(A_-|n\rangle) = (n-1)(A_-|n\rangle). \tag{3.69}$$

Thus the oscillator has precisely the same ladder structure that angular momentum had, except that the ladder now lives in energy rather than in m.

What prevents the ladder from descending forever? Here the answer is positivity:

$$\langle\psi|N|\psi\rangle = \langle\psi|A_+A_-|\psi\rangle = \|A_-|\psi\rangle\|^2 \geq 0. \tag{3.70}$$

So the eigenvalues of N are nonnegative. Since raising and lowering change them in integer steps, there must be a lowest one, and it must be

$$n = 0. \tag{3.71}$$

The corresponding ground state obeys

$$A_-|0\rangle = 0. \tag{3.72}$$

With the standard normalization,

$$A_+|n\rangle = \sqrt{n+1}\,|n+1\rangle, \qquad A_-|n\rangle = \sqrt{n}\,|n-1\rangle, \tag{3.73}$$

and so the full energy spectrum is

$$E_n = \hbar\omega\left(n + \frac{1}{2}\right), \qquad n = 0, 1, 2, \ldots \tag{3.74}$$

This is exactly the spectrum Susskind reaches at the end of the lecture, with the $\frac{1}{2}\hbar\omega$ restored after having been temporarily set aside.

He also remarks, in response to a question, that one does not usually get ladder operators this cleanly. Physics has been lucky here twice: once for angular momentum, and once for the harmonic oscillator. The second luck is especially important because so many systems reduce, near equilibrium, to oscillator behavior.

3.8 Summary

Lecture 3 has a single mathematical instinct running through it. Angular momentum is first reinterpreted as angular dependence of a wavefunction, which leads naturally to the ladder of m-values and to the common multiplet value

$$L^2 = \hbar^2\ell(\ell + 1). \tag{3.75}$$

That algebra then reduces the central-force Schrödinger equation to a radial one-dimensional problem with an effective centrifugal barrier. The resulting spectra are organized by node number and by the $2\ell + 1$ degeneracy of each multiplet, with the Coulomb potential singled out by its special $1, 4, 9, 16, \ldots$ shell pattern before spin is included. The lecture closes by showing that the same algebraic way of thinking works again for the harmonic oscillator: factorize the Hamiltonian, compute a commutator, build a ladder, identify the lowest state, and read off the spectrum.

Chapter 4

Harmonic Oscillator Review, Spin-$\frac{1}{2}$, and the First Exchange Principle

These notes follow Leonard Susskind's fourth lecture in the supplementary *Advanced Quantum Mechanics* sequence, with curation by LazyingArt LLC. The lecture unfolds in a single arc: a purposeful review of the harmonic oscillator as preparation for later field theory, a rapid but concrete return to spin-$\frac{1}{2}$ as angular momentum, an atomic-physics motivation for spin and exclusion, and finally an intentionally unfinished opening of identical-particle statistics. Throughout the oscillator discussion Susskind sets $m = 1$ and mostly suppresses $\hbar$; we follow that convention and only mention $\hbar$ when the lecture does.

4.1 Why the Harmonic Oscillator Returns

Susskind begins by explaining why the harmonic oscillator keeps returning. Any system with an equilibrium configuration, once displaced slightly from equilibrium, will generally oscillate. A violin string is the easy example: it vibrates in a fundamental mode together with higher harmonics. But the real point is not acoustics. The oscillators that matter most for the rest of the course are the oscillatory modes of fields.

This is already visible in electromagnetism. The equilibrium configuration of the electromagnetic field is simply the vacuum: empty space with no electric or magnetic field. If one perturbs that equilibrium, for example by sending radiation into a cavity or waveguide, the field begins to oscillate in its allowed modes. Later those modes will be related to photons. The same pattern reappears in solids: a crystal does not consist of independent atoms oscillating separately, but of coupled atoms whose collective motions propagate as waves. Those oscillatory modes are again described by harmonic oscillators, and their quanta are phonons.

The review is therefore not a detour into familiar material. It is preparation for everything that follows: photons, phonons, and more generally the oscillatory structure that lies behind quantum field theory.

4.2 Oscillator Algebra in Ladder Form

With that motivation in place, Susskind returns to the algebraic description. In the convention $m = 1$, the Hamiltonian is

$$H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 x^2, \tag{4.1}$$

where ω is both the oscillator frequency and the parameter setting the curvature of the potential.

The creation and annihilation operators are defined by

$$a_\pm = \frac{p \pm i\omega x}{\sqrt{2\omega}}. \tag{4.2}$$

The factor $\sqrt{2\omega}$ is not decorative. It is chosen so that the commutator takes the simplest possible form,

$$[a_-, a_+] = 1. \tag{4.3}$$

The number operator is then

$$N = a_+ a_-. \tag{4.4}$$

This commutator, together with the definition of N, implies that a_+ and a_- raise and lower the eigenvalues of N by one unit. The spectrum of N is therefore

$$0, 1, 2, 3, \ldots$$

The lecture also recalls how the Hamiltonian is expressed in terms of N. Classically one would expect the product $a_+ a_-$ simply to reconstruct the Hamiltonian. Quantum mechanically the noncommutativity produces an extra term:

$$\begin{aligned} N &= \frac{1}{2\omega}(p + i\omega x)(p - i\omega x) && (4.5)\\ &= \frac{1}{2\omega}(p^2 + \omega^2 x^2 + i\omega[x, p]) && (4.6)\\ &= \frac{1}{2\omega}(p^2 + \omega^2 x^2 - \omega), && (4.7) \end{aligned}$$

since $[x, p] = i$ in the $\hbar = 1$ convention. Therefore

$$H = \omega\left(N + \frac{1}{2}\right). \tag{4.8}$$

Susskind also writes the simplified form

$$H = \omega N \tag{4.9}$$

when the additive constant is dropped. He is explicit about the distinction: the zero-point term $\frac{1}{2}\omega$ is real, but for many purposes it can be ignored because an additive constant in the energy does not change the equations of motion.

The extra $\frac{1}{2}\omega$ is the ground-state energy of the oscillator, or, in field-theory language, the vacuum energy. Susskind emphasizes that it is a consequence of the uncertainty principle: x and p cannot both vanish simultaneously, so the positive terms in the Hamiltonian cannot both be driven to zero.

4.3 A Student Question, Wave Functions, and the Ground State

At this point a student interrupts with a natural question: why does the ladder include 0 at all? Susskind first answers abstractly, before passing to wave functions. Because the Hamiltonian is built from $p^2 + \omega^2 x^2$, it cannot have arbitrarily negative eigenvalues. There must therefore be a bottom state. The only way the ladder can stop is if the lowering operator annihilates that state:

$$a_-|0\rangle = 0. \tag{4.10}$$

Applying $N = a_+ a_-$ to the same state gives

$$N|0\rangle = 0. \tag{4.11}$$

So the bottom state has number eigenvalue 0. Susskind pauses over the notation: the symbol 0 here is merely the name of the state, not the zero vector.

Only after that abstract argument does he place the wave-function description "parallel" to the operator structure. State vectors become wave functions $\psi(x)$, and energy eigenstates may be written as $\psi_0(x), \psi_1(x), \ldots$. In the position representation,

$$X \mapsto x, \qquad P \mapsto -i\,\frac{d}{dx}. \tag{4.12}$$

If one restores $\hbar$, this becomes $P \mapsto -i\hbar\, d/dx$; Susskind remarks that dimensional analysis tells one where the $\hbar$'s must go.

The time-independent Schrödinger equation is

$$\left[-\frac{1}{2}\frac{d^2}{dx^2} + \frac{1}{2}\omega^2 x^2\right]\psi(x) = E\,\psi(x). \tag{4.13}$$

As he stresses, this equation can be solved formally for any E, but most formal solutions are not physical. For generic energy the wavefunction explodes at infinity and cannot be normalized. The admissible states are square-integrable wavefunctions,

$$\int_{-\infty}^{\infty} \psi^*(x)\psi(x)\,dx = 1. \tag{4.14}$$

Only for special values of E does the differential equation yield such honest eigenfunctions.

The smart route to ψ_0. Rather than solve the second-order equation directly, Susskind uses the simpler first-order equation that must hold in the ground state:

$$a_-\psi_0(x) = 0. \tag{4.15}$$

Since an overall numerical factor is irrelevant in a homogeneous equation, one may write

$$a_- \propto p - i\omega x.$$

Substituting $p = -i\,d/dx$ gives

$$(p - i\omega x)\psi_0(x) = 0 \qquad \Longrightarrow \qquad \left(\frac{d}{dx} + \omega x\right)\psi_0(x) = 0. \tag{4.16}$$

This is much simpler than the full Schrödinger equation. The standard trick is to set

$$\psi_0(x) = e^{f(x)}. \tag{4.17}$$

Then

$$\psi_0'(x) = f'(x)e^{f(x)},$$

so the common factor $e^{f(x)}$ drops out and one is left with

$$\frac{df}{dx} + \omega x = 0. \tag{4.18}$$

Integrating once,

$$f(x) = -\frac{1}{2}\omega x^2 + C. \tag{4.19}$$

Hence the ground-state wavefunction is

$$\psi_0(x) \propto e^{-\omega x^2/2}. \tag{4.20}$$

This is the Gaussian Susskind draws on the board: smooth, concentrated near the origin, and rapidly decaying. It is manifestly square integrable. Its energy is already known from the ladder algebra,

$$E_0 = \frac{\omega}{2}. \tag{4.21}$$

If one wants the normalized form in the lecture convention, it is

$$\psi_0(x) = \left(\frac{\omega}{\pi}\right)^{1/4} e^{-\omega x^2/2}. \tag{4.22}$$

4.4 The First Excited State, Higher States, and the Semiclassical Picture

Once the ground state is explicit, Susskind immediately asks for the first excited state. He again uses the algebra rather than attacking the differential equation afresh:

$$\psi_1(x) \propto a_+\psi_0(x). \tag{4.23}$$

Since $a_+ \propto p + i\omega x$, one has

$$\psi_1(x) \propto (p + i\omega x)\psi_0(x). \tag{4.24}$$

But from the ground-state equation $(p - i\omega x)\psi_0 = 0$, it follows that

$$p\psi_0 = i\omega x\,\psi_0.$$

Therefore

$$(p + i\omega x)\psi_0 = 2i\omega x\,\psi_0, \tag{4.25}$$

and so, up to an irrelevant constant,

$$\psi_1(x) \propto x\,e^{-\omega x^2/2}. \tag{4.26}$$

This is enough to read off the physics. The first excited state is an odd function:

$$\psi_1(-x) = -\psi_1(x).$$

Its probability density is even, but the wavefunction itself changes sign at the origin, and in fact vanishes there. Thus the probability of finding the oscillator exactly at the equilibrium point is zero in the first excited state. The derived form also makes clear that the first excited state has one node, at $x = 0$.

The lecture does not derive a closed formula for all higher states, and it is better not to force one in at this stage. The essential qualitative structure is already visible. Repeated action of a_+ produces higher polynomials multiplying the same Gaussian envelope. Each higher state acquires another node; the parity alternates between even and odd; and the wavefunction is pushed farther outward from the origin. In standard notation these polynomials are Hermite polynomials, but the lecture's emphasis is not the name of the family. It is the picture.

That picture is simple and physical. More support at large $|x|$ means more potential energy. More rapid oscillation means more momentum. As the energy increases, the wavefunction develops more and more "wiggles" and extends farther and farther into the wings. Susskind's point is that the oscillator is trying to do what classical intuition says it should do: high-energy states are likely to be found farther from the center, but when they pass through the center they carry larger momentum.

The lecture then pauses over the correspondence principle. A single high-n eigenstate is not itself a classical oscillator moving back and forth. To see something that looks classical, one must form a wave packet by superposing many energy eigenstates. Susskind's example is a displaced Gaussian: start with the ground-state packet and move it away from the origin. It is no longer an energy eigenstate, so under the time-dependent Schrödinger equation its components pick up different phases. The packet then swings back and forth, becoming more oscillatory near the center and smoother near the turning points, while maintaining its overall character. In that way the harmonic oscillator does satisfy the correspondence principle, but the classical motion is seen in a packet, not in a stationary eigenfunction.

The lecture ends the oscillator discussion with one further remark. Exactly solvable systems are rare. The harmonic oscillator is one of them, and the idealized hydrogen atom is another; both will continue to serve as laboratories for more general ideas.

4.5 Spin-$\frac{1}{2}$ as Angular Momentum

Susskind then explicitly changes topic. The other subjects on his list for the evening are spin and the boson-fermion difference, and spin comes first. The return to spin is not meant as a fresh introduction. It is a reinterpretation of the earlier two-state system in the language of angular momentum.

For a particle at rest there is no orbital angular momentum $\mathbf{R} \times \mathbf{P}$, but there may still be spin. One may picture that spin literally as a tiny object spinning, or one may regard it as an abstract intrinsic property attached to the particle. Susskind is deliberately permissive about the picture. What matters is the mathematics: whatever this extra degree of freedom is, it transforms as angular momentum.

The decisive criterion is the commutation relations. If three operators are attached to the spatial directions x, y, and z, and if they satisfy the angular-momentum algebra, then they are angular

momentum. In particular,

$$[L_z, L_x] = iL_y, \tag{4.27}$$

with the other relations obtained by cyclic permutation.

The candidate operators are the Pauli matrices,

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}. \tag{4.28}$$

Susskind chooses the representation in which σ_z is diagonal. Its eigenvalues are $+1$ and -1, corresponding to spin up and spin down along the z-axis.

The board calculation proceeds by explicit multiplication:

$$\sigma_z\sigma_x = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\sigma_y, \tag{4.29}$$

$$\sigma_x\sigma_z = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = -i\sigma_y. \tag{4.30}$$

Therefore

$$[\sigma_z, \sigma_x] = 2i\sigma_y. \tag{4.31}$$

This is almost the angular-momentum algebra, but not quite. The mismatch is the factor of 2. The fix is exactly the one visible on the board:

$$S_i = \frac{\sigma_i}{2}. \tag{4.32}$$

Then the operators become

$$\frac{\sigma_z}{2} = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}, \qquad \frac{\sigma_x}{2} = \begin{pmatrix} 0 & \frac{1}{2} \\ \frac{1}{2} & 0 \end{pmatrix}, \qquad \frac{\sigma_y}{2} = \begin{pmatrix} 0 & -\frac{i}{2} \\ \frac{i}{2} & 0 \end{pmatrix}, \tag{4.33}$$

and now

$$[S_z, S_x] = iS_y, \tag{4.34}$$

again with cyclic permutations. The two-state system is therefore not merely analogous to angular momentum; it is the simplest nontrivial angular-momentum system.

Its z-component has eigenvalues

$$m_s = \pm\frac{1}{2}. \tag{4.35}$$

This is the spin-$\frac{1}{2}$ doublet. Susskind emphasizes that many of the particles one most naturally thinks of as fundamental — electrons, muons, neutrinos, quarks — carry precisely this kind of spin.

4.6 From Spin to Atoms and Shell Counting

Once the algebraic point is in place, the lecture turns to why nature should care. There are two kinds of angular momentum: orbital angular momentum and spin. Their vector sum is the total angular momentum,

$$\mathbf{J} = \mathbf{L} + \mathbf{S}. \tag{4.36}$$

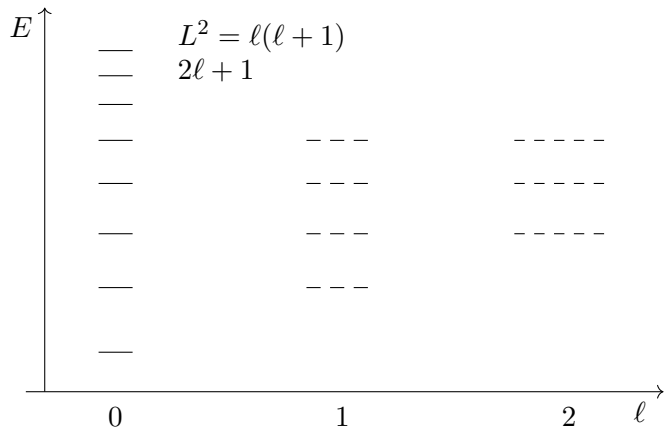


Figure 4.1: Schematic hydrogenic degeneracy pattern, redrawn from the lecture board and transcript. The vertical spacing is not to scale; the important content is the alignment of the $\ell = 1$ and $\ell = 2$ ladders with higher $\ell = 0$ states.

For a particle,

$$\mathbf{L} = \mathbf{R} \times \mathbf{P}.$$

As Susskind later stresses, angular momentum is always defined relative to a chosen origin. In atomic physics the natural origin is the nucleus. The detailed rules for adding $\mathbf{L}$ and $\mathbf{S}$ are postponed; the lecture only needs the notation and the physical distinction.

The atomic motivation begins with the hydrogen atom. In the lecture convention,

$$L^2 = \ell(\ell+1), \tag{4.37}$$

and for fixed ℓ the multiplicity is

$$2\ell + 1. \tag{4.38}$$

Susskind redraws the spectrum schematically as an energy-versus-ℓ plot. The actual hydrogen energies are negative, but for the drawing he simply shifts the picture upward and focuses on degeneracies rather than scale.

The $\ell = 0$ column contains the s-wave tower. The first $\ell = 1$ level is degenerate with the second $\ell = 0$ level, the first $\ell = 2$ level with the third $\ell = 0$ level, and so on. Counting the states shell by shell gives

$$1, \qquad 1+3=4, \qquad 1+3+5=9, \qquad 1+3+5+7=16, \tag{4.39}$$

or more compactly,

$$1, 4, 9, 16, \ldots = n^2. \tag{4.40}$$

This is only the orbital counting. Once spin is included, the idealized shell totals double to

$$2, 8, 18, 32, \ldots = 2n^2. \tag{4.41}$$

The lecture then backs up and turns that counting into a physical argument. Helium is well described, at first approximation, by putting two electrons into the lowest orbital state. But when one proceeds to lithium, the third electron does not drop into the same state. It enters the first excited shell. Why should that happen, when it would seem energetically cheaper to place all the electrons in the lowest orbital?

Pauli's answer is the exclusion principle: no two electrons can occupy the same state. But that raises an immediate problem, because helium plainly allows two electrons in the same orbital. The way out is to recognize that the electron has more structure than its orbital motion alone. It carries an additional two-valued property. In the lecture this is introduced in the simplest possible language: call the two values up and down. Then two electrons may occupy the same orbital state provided they have opposite spin. They are not, after all, in the same full quantum state.

This is the lecture's route to the second row of the periodic table. The first excited shell contains one $\ell = 0$ state and three $\ell = 1$ states, for a total of four spatial states. Doubling by the two spin values gives eight possibilities, matching the eight entries in the second row of the periodic table in this very idealized shell picture.

Susskind also mentions two further points that keep the discussion honest. First, spectroscopy pointed in the same direction as chemistry, although the periodic-table argument is more transparent. Second, once the new two-valued property is interpreted as angular momentum, it ought to reveal itself in a magnetic field. A charged particle carrying angular momentum behaves like a tiny magnet, so the two spin orientations should split in energy when a magnetic field is applied. That splitting is indeed observed.

At the same time, the lecture is explicit that this shell model is naive. It ignores electron-electron interactions and therefore breaks down rather quickly for larger atoms. The simple $2n^2$ pattern is useful here as motivation for spin and exclusion, not as a quantitatively complete many-electron theory.

4.7 Identical Particles and the First Exchange Principle

This brings Susskind to the second advertised subject: bosons and fermions. A student asks whether the exclusion principle is a separate postulate. Susskind's answer is historically layered. In nonrelativistic quantum mechanics it can be stated as a postulate. In relativistic quantum theory it becomes deeper. But for the moment he wants a cleaner quantum-mechanical formulation, one phrased directly in terms of identical particles.

He begins by stepping back to classical statistical mechanics. Suppose two particles of the same kind are placed in boxes. If one imagines that the particles secretly carry names — Harry and Sally is the lecture's example — then placing Harry in box 1 and Sally in box 2 seems different from placing Sally in box 1 and Harry in box 2. One may nevertheless choose to count those as the same configuration, and in classical statistical mechanics that choice leads to the familiar $1/n!$ factors. The point is that classically one can imagine faint labels, even if no experiment can detect them.

Quantum mechanics is stricter. One does not have the luxury of treating truly identical particles as secretly labeled. Swapping two electrons is an honest transformation on the quantum state, not a bookkeeping convention.

For one particle the wavefunction is $\psi(x)$. For two identical particles the state is described by a wavefunction of two arguments,

$$\psi(x_1, x_2), \tag{4.42}$$

where x_1 and x_2 label the two particle slots, not two spatial directions. The probability density $|\psi(x_1, x_2)|^2$ is the probability of finding one particle at x_1 and the other at x_2.

The next move is the key one. Consider the operation that swaps the two particles. Susskind

denotes it by S; to avoid confusion with the spin operator $\mathbf{S}$, we shall write it as P_{12}. Its action is

$$P_{12}\psi(x_1, x_2) = \psi(x_2, x_1). \tag{4.43}$$

A function of two variables need not be symmetric, so in general this produces a new wavefunction.

Now swap twice. Nothing mysterious happens: interchanging the two particle slots a second time returns the original state. Therefore

$$P_{12}^2 = 1. \tag{4.44}$$

Susskind also notes that exchange is a transformation on the Hilbert space and so must be unitary. A unitary operator whose square is the identity can only have eigenvalues

$$\lambda = \pm 1. \tag{4.45}$$

Thus the physically relevant two-particle states fall into two classes:

$$P_{12}\psi = \psi, \qquad \text{or} \qquad P_{12}\psi = -\psi. \tag{4.46}$$

That is exactly where the lecture stops. The wavefunction of two identical particles either returns to itself under exchange or returns with a minus sign. The full interpretation of those two possibilities as bosonic and fermionic statistics is deferred to the next lecture. Susskind is about to begin with the $+1$ case when he announces that he is fading and leaves the rest for next time.

4.8 Summary

The lecture begins with the ubiquity of the harmonic oscillator and with a clear warning that this review is preparation for later field theory, not repetition for its own sake. From there it rebuilds the oscillator algebra, explains the zero-point shift, and answers a student question about why the ladder bottoms out. The abstract argument for a lowest state is then matched by an explicit position-space construction of the Gaussian ground state and the odd first excited state.

Only after that oscillator machinery is in place does the lecture pivot to spin, showing that the Pauli matrices become angular momentum operators once normalized as $S_i = \sigma_i/2$. The atomic discussion then uses hydrogenic degeneracies and naive shell counting to motivate Pauli's extra two-valued electron property, later recognized as spin. Finally, exclusion is reformulated as the first statement of identical-particle exchange: for two identical particles, swapping the particle slots leaves the wavefunction either unchanged or multiplied by -1. That deliberately unfinished dichotomy is the chapter's endpoint and the next lecture's starting point.

Chapter 5

Spin, Exchange, and the Two Minus Signs

These notes follow Leonard Susskind's fifth lecture in the supplementary Advanced Quantum Mechanics sequence, with explicit credit to Leonard Susskind and curation by LazyingArt LLC. The lecture begins in an apparently old-fashioned way, with helium, lithium, and shell filling, because that is where spin and exclusion first show up together in actual matter. Only after that empirical reminder does Susskind pivot to the more abstract point: the Pauli principle is really a statement about many-particle wavefunctions, and its relation to half-integer spin is part of a larger story that he calls a tale of two minus signs.

5.1 Atomic-Shell Motivation for Spin and Exclusion

Susskind opens by recalling that the fermionic character of the electron seems to involve two facts at once. One is that the electron has spin 1/2, so that, forgetting orbital motion for the moment, it has two spin states. The other is the Pauli exclusion principle. In the language of atomic shells, one cannot keep putting electrons into the same state. But, as he stresses immediately, when one talks about a state, spin counts.

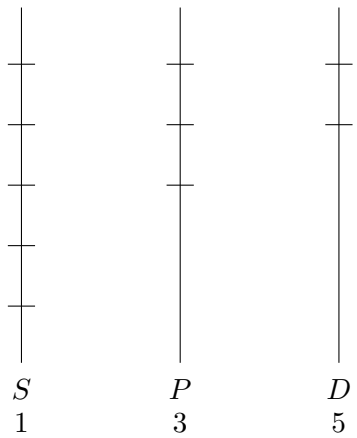
This is why two electrons may occupy the lowest spatial orbital of a helium atom. If spin were ignored, that would look forbidden. But the full one-particle state includes both the spatial wavefunction and the spin label, so one may place two electrons in the lowest orbital provided their spin labels differ. In the crude shell-model language, one is spin up and the other is spin down. Susskind even pauses to say that, more precisely, the two-electron spin state is entangled; the simple up-down language is only a mnemonic.

The lecture is careful about the level of approximation. This is the old shell-model picture, the "chemist's bad approximation," not a modern treatment of interacting electrons. It is being used only because it remembers the right counting. In that simple Coulomb picture, orbital angular momentum l contributes

$$2l + 1 \tag{5.1}$$

spatial states. Thus

$$l = 0 \Rightarrow 2l + 1 = 1, \qquad l = 1 \Rightarrow 2l + 1 = 3, \qquad l = 2 \Rightarrow 2l + 1 = 5. \tag{5.2}$$



These are the familiar S, P, and D sectors.

The visual point of the board sketch is only schematic: there are S-levels, P-levels, and then D-levels, with the Coulomb problem producing extra degeneracies across them. For the counting argument one needs only the first excited shell. It contains one $2s$ orbital and three $2p$ orbitals, so

$$N_{\text{first excited shell, spatial}} = 1_{2s} + 3_{2p} = 4. \tag{5.3}$$

Electrons carry two spin states, and therefore the same shell contains

$$N_{\text{first excited shell, electron}} = 2 \times 4 = 8 \tag{5.4}$$

electron states.

That is the shell-filling logic behind lithium, beryllium, and then neon. The outer electron in lithium must move into the next shell because there is simply no room left in the lowest one. The first excited shell can hold eight electrons because its four spatial states are doubled by spin. After neon the next electron must again go higher, which is why sodium looks chemically like lithium: each has one valence electron outside a closed shell.

Susskind then stops himself. He says, in effect, that he did not want to do chemistry that evening. He wanted only to remind the audience that two things seem to go together in nature: half-integer spin and exclusion. He sharpens the question by contrasting electrons with photons. Photons certainly do not obey exclusion; classical electromagnetic waves are precisely situations in which many, many quanta occupy the same state. In the bosonic case there is no occupancy bound at all. The lecture is therefore led to the real question: why do half-integer spin and exclusion come together, while integer spin and unlimited occupancy come together?

5.2 Many-Particle Wavefunctions and Identical Particles

With that empirical picture in place, Susskind announces the more abstract part of the lecture. The exclusion principle is not just a rule of atomic bookkeeping. It is a statement about the wavefunctions of many-particle systems.

He asks us, for the moment, to forget spin and restore it later. Let x_i denote the position of particle i. Here the subscript i labels the particle, not a Cartesian direction. The three spatial coordinates are being suppressed into a single symbol x.

For a system of n particles, a natural basis is

$$|x_1, x_2, \ldots, x_n\rangle, \tag{5.5}$$

the state in which particle 1 is at x_1, particle 2 is at x_2, and so on. If the abstract many-particle state is $|\Psi\rangle$, then its coordinate-space wavefunction is

$$\psi(x_1, x_2, \ldots, x_n) = \langle x_1, x_2, \ldots, x_n \mid \Psi\rangle. \tag{5.6}$$

The quantity

$$|\psi(x_1, x_2, \ldots, x_n)|^2 \tag{5.7}$$

is the joint probability density for finding particle 1 at x_1, particle 2 at x_2, and so forth.

The lecture board often uses uppercase Ψ for a many-particle wavefunction, and that notation is visible in the later exchange-symmetry sketch. In these notes, to keep the hierarchy clean, $|\Psi\rangle$ denotes the abstract state vector while $\psi(x_1, \ldots, x_n)$ denotes its coordinate representation.

Now comes the new point. Suppose the particles are identical. Then one must ask whether the configuration "particle 1 at x_1, particle 2 at x_2" is physically the same as "particle 1 at x_2, particle 2 at x_1." Susskind dramatizes the question by imagining a blue electron and a red electron, but then immediately retracts the colors: electrons have no such tags. They are indistinguishable. So one is not allowed to build the physics out of bookkeeping labels that nature does not supply.

At the level of state vectors, this already suggests that exchanging two identical particles should not produce a genuinely new physical state. But quantum mechanics admits one further subtlety: overall phase is unobservable. That opens the possibility that exchange may return the same physical state only up to a phase.

5.3 Exchange Phase, Bosons, and Fermions

The transition is sharp. Susskind says this lecture is a tale of two minus signs, and the first of them has to do with exchange. Let us take two identical particles. Since only relative phases matter, the most general possibility consistent with indistinguishability is

$$\psi(x_1, x_2) = e^{i\phi}\, \psi(x_2, x_1). \tag{5.8}$$

Here $e^{i\phi}$ is a fixed phase.

Now exchange the particles twice. After the first swap one has picked up $e^{i\phi}$; after the second swap one has picked up another factor of the same phase. But two swaps bring the configuration back to itself, so

$$\psi(x_1, x_2) = e^{i\phi}\, \psi(x_2, x_1) \tag{5.9}$$

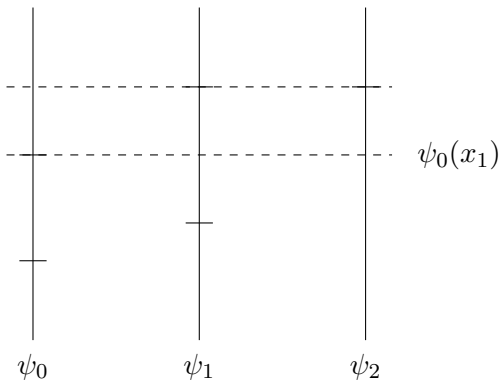
$$= e^{2i\phi}\, \psi(x_1, x_2). \tag{5.10}$$

Therefore

$$e^{2i\phi} = 1. \tag{5.11}$$

In the three-dimensional setting of this lecture, Susskind keeps the two familiar possibilities

$$e^{i\phi} = +1 \qquad \text{or} \qquad e^{i\phi} = -1. \tag{5.12}$$



Definition 5.1. Bosonic wavefunctions are symmetric under exchange:

$$\psi_B(x_1, x_2) = +\psi_B(x_2, x_1). \tag{5.13}$$

Fermionic wavefunctions are antisymmetric under exchange:

$$\psi_F(x_1, x_2) = -\psi_F(x_2, x_1). \tag{5.14}$$

This is the first minus sign of the lecture. A single swap, which one might naively have thought did nothing, changes the sign of a fermionic wavefunction. A double swap is trivial in both cases, as it must be.

Susskind is careful to locate the statement correctly. He explicitly remarks that the story changes in one and two spatial dimensions, so the reduction to ± 1 should be understood here as the three-dimensional story. In ordinary three-dimensional quantum mechanics, however, these are the two sensible possibilities, and they divide identical particles into bosons and fermions.

One further point matters for the lecture's rhythm. The sign itself is not yet observable by itself, because an overall minus sign does not change probabilities or expectation values. Nevertheless it is real in the quantum-mechanical sense that it changes how amplitudes combine. That is why Susskind says the exclusion principle has a deeper meaning than the sentence "you cannot put two particles into the same state." It is first a statement about symmetry of amplitudes.

5.4 Two-Particle Constructions and the Pauli Principle

Susskind now does what he always does when a formal rule threatens to become too abstract: he tests it on explicit states. The board picture at this stage shows a set of candidate one-particle orbitals, labeled ψ_0, ψ_1, ψ_2, with ψ_0 the lowest one. The point of the figure is not an accurate spectrum, but a clean bookkeeping device for discussing occupation.

A one-particle orbital becomes, in a two-particle wavefunction, a factor such as $\psi_0(x_1)$. With that understood, Susskind studies three cases.

Two particles in the same one-particle state. Start with

$$\psi_0(x_1)\psi_0(x_2). \tag{5.15}$$

Under exchange $x_1 \leftrightarrow x_2$, nothing changes:

$$\psi_0(x_2)\psi_0(x_1) = \psi_0(x_1)\psi_0(x_2). \tag{5.16}$$

The reason is elementary but important: wavefunctions are numerical functions, not operators, so the factors commute. This state is therefore symmetric. It is perfectly acceptable for bosons, but it cannot be a fermionic state.

Two particles in different one-particle states. Now try

$$\psi_0(x_1)\psi_1(x_2). \tag{5.17}$$

After exchange one gets

$$\psi_0(x_2)\psi_1(x_1). \tag{5.18}$$

In general the exchanged expression is neither equal to the original product nor equal to its negative. So the bare product is not, by itself, an acceptable bosonic or fermionic wavefunction. This does not mean that one cannot place one particle in ψ_0 and another in ψ_1. It means only that the correct two-particle state must respect the exchange rule.

The proper constructions are the symmetric and antisymmetric combinations:

$$\psi_B^{(01)}(x_1, x_2) = \frac{1}{\sqrt{2}}\Big[\psi_0(x_1)\psi_1(x_2) + \psi_0(x_2)\psi_1(x_1)\Big], \tag{5.19}$$

$$\psi_F^{(01)}(x_1, x_2) = \frac{1}{\sqrt{2}}\Big[\psi_0(x_1)\psi_1(x_2) - \psi_0(x_2)\psi_1(x_1)\Big]. \tag{5.20}$$

The first is unchanged by exchange and is therefore a bosonic state. The second changes sign and is therefore a fermionic state.

The special case that becomes exclusion. Now make the two one-particle states coincide. For bosons one obtains

$$\psi_B^{(00)}(x_1, x_2) \propto \psi_0(x_1)\psi_0(x_2) + \psi_0(x_2)\psi_0(x_1) = 2\,\psi_0(x_1)\psi_0(x_2), \tag{5.21}$$

which is nonzero and can be normalized in the usual way. Two bosons can occupy the same one-particle state.

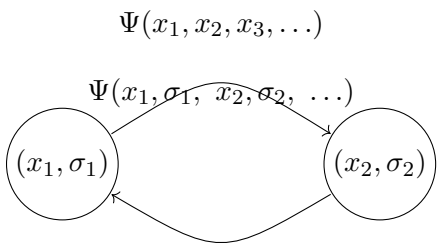
For fermions one instead obtains

$$\psi_F^{(00)}(x_1, x_2) \propto \psi_0(x_1)\psi_0(x_2) - \psi_0(x_2)\psi_0(x_1) = 0. \tag{5.22}$$

This is the lecture's direct mathematical statement of Pauli exclusion. The exclusion principle is not being appended as an extra axiom. It is the statement that the antisymmetric same-state amplitude cancels itself.

This is why Susskind says the exclusion principle can even be viewed as a kind of interference effect. The two exchanged contributions cancel destructively. He does not make that viewpoint the foundation, but he does accept it as a useful interpretation of why the amplitude vanishes.

The same logic extends immediately to many particles. A many-particle bosonic wavefunction is unchanged by any transposition of arguments, while a fermionic one changes sign under any transposition. Double exchange is always trivial; single exchange is not. That is the mathematical structure underlying the exclusion principle.



5.5 Including Spin, Permutations, and the Meaning of Exchange

Only after the spatial story is clear does Susskind restore spin. He asks us to make the smallest possible change: instead of a one-particle wavefunction depending only on position, let it depend on position and spin,

$$\psi(x, \sigma). \tag{5.23}$$

The discrete label σ may be taken, in the lecture, to mean the chosen σ_z value. One may think of $\sigma = \pm 1$, or equivalently $\uparrow, \downarrow$. The quantity

$$|\psi(x, \sigma)|^2 \tag{5.24}$$

is then the probability density to find the particle at x with the corresponding spin value.

For many particles the wavefunction becomes a function of the full labels,

$$\psi(x_1, \sigma_1;\, x_2, \sigma_2;\, \ldots). \tag{5.25}$$

The board picture at this point makes the pedagogical point visually: once spin is included, exchange means exchanging everything associated with particle 1 and particle 2, not merely the positions.

Thus the exchange rule becomes

$$\psi_B(x_1, \sigma_1;\, x_2, \sigma_2) = +\psi_B(x_2, \sigma_2;\, x_1, \sigma_1), \tag{5.26}$$

$$\psi_F(x_1, \sigma_1;\, x_2, \sigma_2) = -\psi_F(x_2, \sigma_2;\, x_1, \sigma_1). \tag{5.27}$$

What follows is exactly the statement with which the lecture began: two fermions cannot occupy the same full state, including spin. Two electrons may therefore share the same spatial orbital only when their spin labels differ.

A student remarks that this sounds like the permutation group, and Susskind agrees completely. Exchange is a permutation, and the ± 1 behavior furnishes a one-dimensional representation of the permutation group. In a compact operator notation one may write

$$P_{12}\psi_\pm = \pm\psi_\pm, \tag{5.28}$$

but the lecture's real point is the explicit one: exchange acts on the entire particle label.

Another student asks about the difference between adding wavefunctions and multiplying them. Susskind uses the question to separate two ideas that are often muddled. If one adds one-particle wavefunctions,

$$\alpha\,\psi_0 + \beta\,\psi_1, \tag{5.29}$$

one is describing a single particle in a superposition of two states. If instead one writes

$$\psi_0(x_1)\psi_1(x_2), \tag{5.30}$$

one is building a two-particle product state: particle 1 in one state, particle 2 in the other. Symmetrization or antisymmetrization comes only after that product has been formed.

The student discussion also foreshadows something Susskind will return to later. Because the same-state fermionic construction vanishes by cancellation of amplitudes, the Pauli principle can be viewed as a peculiar interference phenomenon. At this stage he leaves that remark hanging and turns to the second minus sign.

5.6 Rotation by 2π and the Second Minus Sign

The first minus sign concerned exchange of two identical fermions. The second concerns a single particle with half-integer angular momentum. Susskind asks what happens if one rotates a state by an angle θ, in particular by a full 2π.

At first sight one expects nothing at all. A 2π rotation is geometrically the identity rotation. But the same loophole that appeared in the exchange discussion appears here too: if all that must remain unchanged are probabilities and expectation values, the state may return only up to a phase.

Let $\psi(\theta)$ denote the state after a rotation by angle θ about the z-axis. The generator of such rotations is J_z. In Susskind's convention,

$$J_z\,\psi(\theta) = -i\,\frac{\partial}{\partial\theta}\psi(\theta). \tag{5.31}$$

Now suppose the state is a J_z-eigenstate with eigenvalue m:

$$J_z\,\psi = m\,\psi. \tag{5.32}$$

Combining the two equations gives

$$-i\,\frac{\partial}{\partial\theta}\psi(\theta) = m\,\psi(\theta), \tag{5.33}$$

whose solution is

$$\psi(\theta) = e^{im\theta}\psi(0). \tag{5.34}$$

Therefore

$$\psi(2\pi) = e^{i2\pi m}\psi(0). \tag{5.35}$$

At this point the lecture appeals to the familiar spectrum of angular momentum. The eigenvalues of J_z are either integers or half-integers. Hence

$$m \in \mathbb{Z} \quad\Rightarrow\quad e^{i2\pi m} = +1, \tag{5.36}$$

while

$$m \in \mathbb{Z} + \frac{1}{2} \quad\Rightarrow\quad e^{i2\pi m} = -1. \tag{5.37}$$

Integer-spin states therefore return to themselves under 2π, while half-integer-spin states return with a minus sign. A 4π rotation restores all of them:

$$\psi(4\pi) = \psi(0). \tag{5.38}$$

This is the second minus sign of the lecture. It has nothing directly to do with exchanging two particles. It belongs already to the rotational behavior of a single half-integer-spin state.

Remark 5.2. For notational cleanliness one may also write the rotation operator in the standard form

$$R_z(\theta) = e^{-i\theta J_z}. \tag{5.39}$$

The lecture itself proceeds instead through the differential relation $J_z = -i\,\partial/\partial\theta$ acting on $\psi(\theta)$. Because of this, one should not overemphasize the sign convention in the exponent. The physical invariant is the behavior under 2π: integer spin gives $+1$, half-integer spin gives -1, and 4π gives the identity.

Susskind immediately adds an important warning. The minus sign acquired under a 2π rotation is not itself visible in ordinary probabilities, any more than the exchange sign was. But that does not mean it is mathematically trivial. It is not the identity operation for fermions, only an operation whose effect disappears in $|\psi|^2$.

5.7 Topology, Spin-Statistics, and Observability

At this stage the lecture becomes more heuristic. Susskind shifts from algebra to pictures, demonstrations, and topology. He is not giving Pauli's rigorous proof of the spin-statistics theorem. He is trying to show why the two minus signs do not merely resemble one another accidentally.

The 4π topology. The cup trick and the belt trick are used to exhibit a basic fact about the topology of the rotation group. A path in rotation space corresponding to a 2π rotation is nontrivial; a path corresponding to 4π can be unwound. In Susskind's language, a 2π twist gives a tangle, while after another 2π the tangle can be undone. This is why the mathematics naturally distinguishes 2π from 4π.

The lesson is not yet spin-statistics. It is only that the half-integer behavior under 2π is not a cheap artifact of notation. It reflects the topology of the rotation group in three dimensions.

Exchange plus 2π rotation. The lecture then takes a more suggestive step. Susskind attributes to David Finkelstein a deeper topological observation: exchange of two particles and a 2π rotation of one particle are related. The belt demonstration makes this vivid. A belt configuration that can be read one way as an exchange can be read another way as a 2π twist, and "exchange plus a 2π rotation" becomes topologically trivial.

He then translates the picture into a worldline-type argument. Begin with no particles. Create two particle-antiparticle pairs, so that at an intermediate time there are four particles present. Now rotate one of them by 2π and simultaneously exchange two of the others. Topologically, Susskind argues, this history can be deformed into a trivial creation-annihilation process. A trivial process should carry no net phase. Therefore the phase acquired from exchange must cancel the phase acquired from rotation.

That is the lecture's heuristic version of the spin-statistics link. It is not presented as a proof, and Susskind explicitly says he has never seen it turned into a fully rigorous derivation in the form he likes. But it captures the right moral: the exchange sign and the 2π rotation sign are deeply connected, not merely parallel curiosities.

This is also why he broadens the discussion beyond elementary relativistic particles. Once the real criterion is the ability to create and annihilate particle-like excitations, the argument should

apply equally to solitons, solitary waves, monopoles, and other lump-like excitations in field theory or condensed matter systems. In that sense he takes the topology to be closer to the root of spin-statistics than the specific historical proof.

Why a global sign is invisible. Susskind then asks the operational question: can one observe the minus sign from a 2π rotation? If one rotates a single trapped electron adiabatically through 2π and then compares it with an otherwise identical experiment in which no such rotation was performed, the answer is no. The phase change is only global. Since probabilities depend on $\psi\psi^*$, the sign disappears.

This is the null experiment with which he begins the final part of the lecture. It is important because it prevents confusion. A universal 2π rotation of everything is not what one measures. The measurable effect appears only when one compares two coherent branches relative to one another.

Relative phase from branch interference. Now take one electron and split its wavefunction coherently into two branches. In the lecture these are called ψ_1 and ψ_2; to avoid conflict with the earlier one-particle orbitals, let us write them as ψ_{L} and ψ_{R}:

$$\psi(x) = \psi_{\mathrm{L}}(x) + \psi_{\mathrm{R}}(x). \tag{5.40}$$

This is still one electron. One branch of its wavefunction goes one way, the other branch another way. If one looks to determine which branch the electron took, the interference is destroyed; that is precisely the ordinary two-slit logic that Susskind repeatedly invokes.

Now let only one branch execute a 2π spin rotation, while the other branch is left alone. Then the state becomes

$$\psi(x) \longrightarrow \psi_{\mathrm{L}}(x) - \psi_{\mathrm{R}}(x). \tag{5.41}$$

This minus sign is no longer global. It is a relative phase between coherent branches, and therefore it changes the interference pattern when the branches are brought back together. Constructive interference can become destructive interference, and a point of reinforcement can become a node.

That is the core observability argument. A 2π rotation is unobservable as a global phase, but observable as a relative phase. Susskind then describes a more realistic version of the experiment: one sends a coherent electron beam through two arms, arranges a magnetic field in only one arm, and uses spin precession to make the spin in that arm execute one 2π rotation while the other arm does not. The pure adiabatic version and the practical precession-based version differ in apparatus, but not in conceptual content.

He closes this part with a subtlety. What is being detected is not a literal 2π rotation of the whole world. If the entire universe were rotated, no relative phase would be created and nothing observable would happen. The measurement works only because one branch of the world is rotated relative to another.

An exchange-interference coda. Right at the end, Susskind sketches a two-electron analog. One could imagine coherently splitting a pair of electrons into two branches, exchanging the electrons in one branch but not in the other, and then recombining the branches. The exchange sign would then appear as a relative phase, just as the 2π rotation sign did in the one-electron interferometer. He even suggests a hybrid experiment in which exchange occurs on one branch and a 2π rotation on the other, so that the two effects should cancel. The discussion remains exploratory, and he

does not claim that such an experiment has been carried out. But it reinforces the final conceptual point: quantum indistinguishability is subtler than classical sameness, and the exchange sign is a real ingredient of interference.

5.8 Summary

The lecture proceeds deliberately from chemistry to abstraction. It begins by recalling that, in the old shell picture, state includes spin, so two electrons can share the lowest spatial orbital only when their spin labels differ. That empirical reminder sets up the real subject: the exclusion principle is encoded in the symmetry of many-particle wavefunctions. Bosons have symmetric wavefunctions, fermions antisymmetric ones, and the Pauli principle appears because the antisymmetric same-state combination vanishes. After spin is restored as part of the particle label, the same rule says that two fermions cannot occupy the same full state, including spin. Only then does Susskind introduce the second minus sign, the -1 acquired by half-integer-spin states under a 2π rotation. The final topological and interferometric arguments are deliberately suggestive rather than formal, but they leave the intended impression: exchange and 2π rotation are deeply linked, and the sign picked up by a fermion under 2π becomes physically visible whenever it is converted from a global phase into a relative one.

Chapter 6

Oscillators, Occupation Numbers, and the First Quantum Field

These notes follow Leonard Susskind's sixth lecture in *Advanced Quantum Mechanics*, curated by LazyingArt LLC. The lecture begins at the largest possible scale: quantum field theory is presented as the framework for essentially all known non-gravitational physics. But Susskind immediately narrows the task. Exact quantum field theories are usually too complicated to solve directly, so the evening's entry point is deliberately elementary. Before second quantization can mean anything, the harmonic oscillator has to be put back on the table, stripped of its mechanical ornament, and understood as algebra. From there the lecture unfolds in a definite sequence: one oscillator, many oscillators, occupation numbers, one-particle states in a box, and finally the operator-valued field $\Psi(x)$ that creates and annihilates bosons at a point.

6.1 Why Harmonic Oscillators Come First

Susskind's first move is to postpone the advertised topic. Yes, the lecture is about quantum field theory and second quantization, but before either of those ideas can be useful one must review the harmonic oscillator and generalize it a little. The reason is not that springs are physically fundamental. Quite the contrary: he explicitly asks us not to think about springs very much. What matters is the algebra of the two ladder operators and the number operator built from them. That algebra is the central mathematical structure that later becomes particle creation and annihilation.

The lecture begins with the notation A_+ and A_-, but the same operators are more naturally written as

$$A_+ \to a^\dagger, \qquad A_- \to a.$$

We will use the dagger notation from the outset. The operator $a^\dagger$ raises the excitation level, a lowers it, and their product is the number operator. Later these same operators will literally create and annihilate particles. At the start, however, Susskind wants the class to see only the abstract algebra and its consequences.

6.2 Single-Oscillator Algebra and Energy Quanta

For a single oscillator the lecture starts from the familiar rules

$$a|0\rangle = 0, \tag{6.1}$$

$$N = a^\dagger a, \tag{6.2}$$

$$N|n\rangle = n|n\rangle, \tag{6.3}$$

$$[a, a^\dagger] = 1. \tag{6.4}$$

The state $|0\rangle$ is the bottom state, and the states $|n\rangle$ are eigenstates of the number operator N with eigenvalue n. The operators a and $a^\dagger$ are Hermitian conjugates of one another.

Susskind briefly reminds the audience that these operators may be written in terms of x and p, but the spoken formula is self-corrected on the board and plays no real role in what follows. The lecture quickly settles on the algebraic viewpoint: once one knows a, $a^\dagger$, $N = a^\dagger a$, and $[a, a^\dagger] = 1$, one has essentially everything needed from the oscillator.

The frequency ω survives in the Hamiltonian only as the energy cost per quantum. Up to the zero-point term,

$$H = \hbar\omega \left(N + \frac{1}{2} \right). \tag{6.5}$$

Throughout this lecture the additive constant $\frac{1}{2}\hbar\omega$ is set aside, so the working Hamiltonian is

$$H = \hbar\omega N. \tag{6.6}$$

This is the point at which ω really matters in the present discussion. Raising the oscillator by one level costs one unit of energy $\hbar\omega$; lowering it returns that much energy. The harmonic oscillator is thus already a bookkeeping system for excitations counted by the integer n.

6.3 Many Oscillators and Occupation Numbers

The next step is the minimal generalization that will later open the door to fields. Suppose there are many oscillators, labeled by an index i, possibly infinitely many of them. Susskind's examples are deliberately simple: a collection of springs, or the infinitely many normal modes of an idealized violin string. The important point is that these are independent systems. Once that is assumed, the algebra is forced.

If two subsystems are genuinely independent, operators belonging to one must commute with operators belonging to the other. Otherwise they could not be measured independently. This gives the multimode commutation relations

$$[a_i, a_j] = 0, \qquad [a_i^\dagger, a_j^\dagger] = 0, \qquad [a_i, a_j^\dagger] = \delta_{ij}. \tag{6.7}$$

The last relation contains the whole logic: if $i \neq j$, the oscillators are different and the commutator vanishes; if $i = j$, one recovers the single-oscillator commutator.

Each mode has its own number operator

$$N_i = a_i^\dagger a_i, \tag{6.8}$$

and the total Hamiltonian is the sum of the energies of all modes:

$$H = \sum_i \hbar\omega_i N_i. \tag{6.9}$$

Each oscillator may have its own frequency ω_i. Susskind then introduces the phrase *occupation number* for the eigenvalue n_i of N_i. At first it is only a name, but it is a carefully chosen name: later it will literally count how many particles occupy a given one-particle state.

A basis of states is obtained by listing the occupation number of every oscillator,

$$|n_1, n_2, n_3, \ldots\rangle. \tag{6.10}$$

For a single oscillator this reduces to the usual sequence $|0\rangle, |1\rangle, |2\rangle, \ldots$. For many oscillators it becomes a long list, finite or infinite, of nonnegative integers. The special state with no excitation anywhere is

$$|0\rangle = |0, 0, 0, \ldots\rangle, \tag{6.11}$$

and it is annihilated by every lowering operator:

$$a_i|0\rangle = 0 \qquad \text{for all } i. \tag{6.12}$$

6.4 Deriving the Square-Root Factors

At this point Susskind slows down. He does not want the ladder rules to look arbitrary, so he stops and proves the coefficient in the action of $a^\dagger$. This pause is essential to the lecture. It turns a formal symbol into a controlled piece of mathematics.

States with different eigenvalues of N are orthogonal, and the lecture assumes the standard normalization

$$\langle n|m\rangle = \delta_{nm}. \tag{6.13}$$

With that choice, $a^\dagger|n\rangle$ cannot simply be $|n+1\rangle$; there must be a coefficient. The board writes

$$a^\dagger|n\rangle = c_n|n+1\rangle. \tag{6.14}$$

The constant c_n is to be found from the commutator.

Take the Hermitian conjugate:

$$\langle n|a = c_n\langle n+1|, \tag{6.15}$$

where, as in the lecture, we choose c_n to be real. Now take the inner product of the bra equation with the ket equation:

$$\langle n|aa^\dagger|n\rangle = c_n^2\langle n+1|n+1\rangle = c_n^2. \tag{6.16}$$

The states are normalized, so the right-hand side is simply c_n^2. On the left, use

$$aa^\dagger = a^\dagger a + 1 = N + 1, \tag{6.17}$$

which is the clean version of the commutator identity Susskind reaches after a brief hesitation at the board. Therefore

$$c_n^2 = \langle n|(N+1)|n\rangle = n + 1, \tag{6.18}$$

and hence

$$c_n = \sqrt{n+1}. \tag{6.19}$$

So the raising operator acts as

$$a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle. \tag{6.20}$$

The lowering rule is obtained by the same logic, now starting from

$$a|n\rangle = d_n|n-1\rangle.$$

Since $a^\dagger a = N$,

$$n = \langle n|N|n\rangle = \langle n|a^\dagger a|n\rangle = d_n^2, \tag{6.21}$$

and therefore

$$a|n\rangle = \sqrt{n}\,|n-1\rangle. \tag{6.22}$$

At $n = 0$ this automatically gives $a|0\rangle = 0$, so the formula already encodes the fact that the lowering operator annihilates the bottom state.

Susskind then lifts the result to the multimode occupation-number basis. One acts on the i-th slot exactly as for a single oscillator, leaving all other slots untouched:

$$a_i^\dagger|n_1,\ldots,n_i,\ldots\rangle = \sqrt{n_i+1}\,|n_1,\ldots,n_i+1,\ldots\rangle, \tag{6.23}$$

$$a_i|n_1,\ldots,n_i,\ldots\rangle = \sqrt{n_i}\,|n_1,\ldots,n_i-1,\ldots\rangle. \tag{6.24}$$

This is the content of the partially visible board formula: only the i-th occupation number changes, and the familiar square root belongs to that slot alone. All other occupation numbers are merely bystanders.

A later question in the lecture asks what the square root $\sqrt{n+1}$ "means physically." Susskind's answer is revealing. Sometimes mathematics is just mathematics: the square root is what is needed for the states to remain normalized and for $a^\dagger a$ really to act as the number operator. That is enough.

6.5 From Wavefunctions to Bosonic Fock Space

Only after the oscillator interlude is complete does Susskind return to the promised topic. But he does not define the field immediately. He first explains what the familiar wavefunction ψ is not.

For one particle, $\psi(x)$ is the state vector in the position representation. It is not itself an observable. One does not measure the wavefunction directly; one measures position, momentum, angular momentum, and so forth. If there are several particles, then the wavefunction is not a function of one coordinate but of all particle coordinates:

$$\psi(x_1,\ldots,x_N). \tag{6.25}$$

And that wavefunction describes a fixed number of particles. A fifteen-particle wavefunction describes fifteen particles, not fourteen, not sixteen. This, Susskind says, is *not* the idea of a quantum field.

To build the bridge he says, in effect, "let's go back" to ordinary one-particle quantum mechanics. The example is a particle in a one-dimensional box. The energy eigenstates are a family of one-particle wavefunctions

$$\psi_1(x),\ \psi_2(x),\ \psi_3(x),\ \ldots,\ \psi_i(x), \tag{6.26}$$

ordered by energy. The first is the smoothest function that fits in the box, the next has one node, and higher states oscillate more rapidly. If one wishes to keep a standard model in mind, the ideal infinite well would use

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \qquad 0 < x < L, \tag{6.27}$$

but the lecture itself relies only on the qualitative ordering by increasing node count and energy.

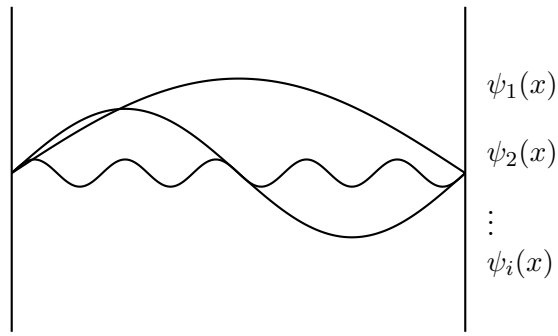


Figure 6.1: Qualitative one-particle box modes, redrawn from the lecture sketch. What matters here is the ordering of modes and their increasing oscillation, not precise amplitudes.

Now the earlier oscillator language suddenly reappears. Suppose there are many bosons, of not necessarily fixed total number, occupying these one-particle states. Then a basis of many-body states is obtained by specifying how many particles occupy each one-particle mode:

$$|n_1, n_2, n_3, \ldots\rangle. \tag{6.28}$$

The resemblance to the many-oscillator basis is not accidental. It is the central structural observation of the lecture. Each one-particle state behaves as though it were its own oscillator, and the occupation number of that oscillator is literally the number of bosons occupying that state.

This is also where the creation and annihilation operators acquire their particle interpretation. Acting with $a_i^\dagger$ increases the number of particles in the i-th one-particle state by one; acting with a_i decreases it by one. The same oscillator algebra is now particle bookkeeping.

The free many-body Hamiltonian is therefore

$$H = \sum_i \omega_i a_i^\dagger a_i. \tag{6.29}$$

Here ω_i is the energy of a particle in the i-th one-particle state, after Susskind has dropped $\hbar$ from the notation. The question period after the break settles several possible confusions about this formula.

First, the ω_i need not be equally spaced. The energy levels of a particle in a box are not harmonic-oscillator levels. What is evenly spaced is something else: for a fixed mode i, the sequence of occupation numbers $0, 1, 2, \ldots$ forms the equally spaced ladder associated with that oscillator.

Second, the particles are free. That is why the total energy is additive. If the particles interacted, one would have to include interaction energy and the Hamiltonian would not be just a sum of one-particle contributions.

Third, the omitted zero-point constant really is irrelevant at this stage. Adding a constant C to the Hamiltonian changes no commutator:

$$[O, H + C] = [O, H]. \tag{6.30}$$

So the equations of motion are unchanged, and only energy differences matter in the present discussion.

The variable-particle-number space spanned by the states $|n_1, n_2, \ldots\rangle$ is what the lecture identifies as Fock space. It is this space, not the fixed-N wavefunction space, on which the quantum field will act.

6.6 The Nonrelativistic Field Operator and What It Does

Only now, after all of these preparations, does Susskind give the definition. He explicitly warns the class not to demand too much intuition too early. Definitions are judged by what they let one do. In that spirit the nonrelativistic bosonic field is introduced as

$$\Psi(x) = \sum_i a_i \, \psi_i(x), \tag{6.31}$$

with Hermitian conjugate

$$\Psi^\dagger(x) = \sum_i a_i^\dagger \, \psi_i^*(x). \tag{6.32}$$

The $\psi_i(x)$ are ordinary one-particle wavefunctions, just numbers once x is specified. The operator character lives entirely in the a_i and $a_i^\dagger$. Thus for each spatial point x there is an operator $\Psi(x)$ acting on Fock space.

This is also where the earlier question about Fourier analysis finds its answer. If the one-particle basis is taken to be momentum eigenstates, then the $\psi_i(x)$ are plane waves or trigonometric modes, and the operators a_i play the role of quantum Fourier coefficients. More generally, the field is expanded in whatever complete orthonormal basis of one-particle states is convenient.

Susskind first motivates Ψ as the kind of quantity one would like to regard as observable. A few minutes later he corrects the point himself: Ψ and $\Psi^\dagger$ are not Hermitian, so taken separately they are not observables in the strict quantum-mechanical sense. The mathematically careful statement is that Ψ is an operator-valued field, and Hermitian combinations such as $\Psi + \Psi^\dagger$, or densities built from them, are the observable quantities. The main point of the lecture, however, is not this distinction but what the field does.

6.6.1 Action on the vacuum

The first calculation is the one Susskind wants the audience to remember. What does $\Psi^\dagger(x)$ do to the vacuum?

Let $|i\rangle$ denote the one-particle state with wavefunction $\psi_i(x)$, so that

$$\langle x|i\rangle = \psi_i(x). \tag{6.33}$$

Because the $|i\rangle$ form a complete basis of one-particle states,

$$\sum_i |i\rangle\langle i| = 1. \tag{6.34}$$

Apply this identity operator to the position eigenstate $|x\rangle$:

$$|x\rangle = \sum_i |i\rangle\langle i|x\rangle \tag{6.35}$$

$$= \sum_i \psi_i^*(x)\,|i\rangle. \tag{6.36}$$

Now rewrite the one-particle state $|i\rangle$ in Fock-space language. It is created from the vacuum by

$$|i\rangle = a_i^\dagger|0\rangle. \tag{6.37}$$

Substituting,

$$|x\rangle = \sum_i \psi_i^*(x)\,a_i^\dagger|0\rangle \tag{6.38}$$

$$= \Psi^\dagger(x)|0\rangle. \tag{6.39}$$

This is the payoff of the definition:

$$\Psi^\dagger(x)|0\rangle = |x\rangle. \tag{6.40}$$

The field operator $\Psi^\dagger(x)$ creates a particle at position x.

By contrast,

$$\Psi(x)|0\rangle = 0, \tag{6.41}$$

because every term in $\Psi(x)$ contains an annihilation operator. The field $\Psi(x)$ removes a particle at x, and if there is none to remove, the result is the zero vector.

6.6.2 Vacuum versus zero

Susskind dwells on this point because the notation is treacherous. The vacuum $|0\rangle$ is not the zero vector. It is a normalized state, of norm one, representing empty space in this simplified theory. The symbol 0 in $|0\rangle$ means zero particles, not zero length.

When an annihilation operator acts on the vacuum, the result is the genuine zero vector,

$$a_i|0\rangle = 0. \tag{6.42}$$

That vector has zero norm and cannot be normalized. It does not describe a physical state. The lecture spends time emphasizing this because the distinction matters. A superposition of the vacuum with a one-particle state is a meaningful state with indefinite particle number. Adding the zero vector to a state changes nothing at all.

6.6.3 Two particles and bosonic symmetry

Once $\Psi^\dagger(x)$ is understood as "put a particle at x," the two-particle state is obtained by acting again:

$$\Psi^\dagger(y)\Psi^\dagger(x)|0\rangle. \tag{6.43}$$

Expanding both fields gives

$$\Psi^\dagger(y)\Psi^\dagger(x) = \sum_{i,j} a_i^\dagger a_j^\dagger\,\psi_i^*(y)\psi_j^*(x). \tag{6.44}$$

But all creation operators commute,

$$[a_i^\dagger, a_j^\dagger] = 0, \tag{6.45}$$

so the field operators commute as well:

$$[\Psi^\dagger(x), \Psi^\dagger(y)] = 0. \tag{6.46}$$

Therefore

$$\Psi^\dagger(y)\Psi^\dagger(x)|0\rangle = \Psi^\dagger(x)\Psi^\dagger(y)|0\rangle. \tag{6.47}$$

This is exactly the symmetry of bosons under exchange. In the lecture Susskind points out that one could almost reverse the logic: had one started with this construction alone, the symmetry would have shown that the particles produced in this way are bosons.

A single field Ψ describes a single species of boson. Different bosonic species require different fields. The spatial argument tells the field where to create or annihilate the particle; the field itself tells us what kind of particle it is.

6.6.4 Why this formalism is useful

The lecture closes by returning to motivation. If one were studying a world with exactly thirty-seven particles, never thirty-six and never thirty-eight, then this machinery would be an unnecessarily elaborate detour. But quantum field theory is designed for a different problem: systems in which the particle number may vary, and even states that are superpositions of different particle numbers. In that sense it is, as Susskind says, a bookkeeping device, though an extremely powerful one.

In particular, the formalism naturally describes states built from

$$|0\rangle, \qquad \Psi^\dagger(x)|0\rangle, \qquad \Psi^\dagger(y)\Psi^\dagger(x)|0\rangle, \qquad \text{and so on,}$$

and also superpositions of sectors with different numbers of particles. A laser field is the lecture's example: not a state of a definite number of photons, but a superposition of many photon-number sectors. That is precisely the sort of situation for which the field-operator language is meant.

6.7 Summary

The lecture enters quantum field theory by a deliberate sequence of reductions and reinterpretations. First, the harmonic oscillator is recast as pure operator algebra. Then many independent oscillators are organized by occupation numbers. Next, a one-particle basis in a box is reinterpreted as a family of modes that can each be occupied by any number of bosons. Only after that preparation does the field appear:

$$\Psi(x) = \sum_i a_i \psi_i(x), \qquad \Psi^\dagger(x) = \sum_i a_i^\dagger \psi_i^*(x). \tag{6.48}$$

The relation between particles and fields is then no longer mysterious. The operator $\Psi^\dagger(x)$ acting on the vacuum creates a particle at x, repeated action creates multiparticle states, and the commutativity of the creation operators makes those particles bosons. What began as oscillator algebra has become a formalism for states with indefinite particle number. That is the lecture's central lesson: in this simplest nonrelativistic setting, quantum field theory is the natural language for counting, creating, annihilating, and superposing quanta.

Chapter 7

From Particles to a Simple Quantum Field

These notes follow Leonard Susskind's seventh lecture in *Advanced Quantum Mechanics*. The lecture is deliberately a second pass through the construction of a simple quantum field: not quite a repetition, but the same idea rebuilt more slowly and from a slightly different angle. Susskind starts from particles, not from classical fields, and asks what formalism can describe one particle, two particles, or any number of particles, including situations in which the number changes. The answer is a nonrelativistic field theory for a single species of particle. The exposition below keeps that order, that pacing, and that motivation, with curation by LazyingArt LLC.

7.1 Why One Needs Fields

Susskind opens by reminding the audience that there are two routes into the subject. One can begin with particles and understand why they are naturally described by fields, or one can begin with fields and understand why wave fields come in quanta. This lecture stays with the first route.

The motivation is physical before it is formal. Fixed-particle-number quantum mechanics is perfectly adequate if one insists on studying one particle, or two particles, or any other predetermined number. But the world does not work that way. Photons are emitted and absorbed continually. Unstable particles decay. A neutron can decay into a proton, an electron, and a neutrino, and in the process the particle content changes. If quantum mechanics is to accommodate such processes within a single framework, it must allow the number of particles to vary.

The lecture therefore narrows the scope in order to sharpen the idea. There is one species of nonrelativistic particle, and one wants a formalism broad enough to describe all occupation numbers of that species at once. Susskind also warns at the outset that the mathematics is not hard, but it is abstract; this is why he restarts the construction from the one-particle theory and rebuilds the necessary pieces carefully.

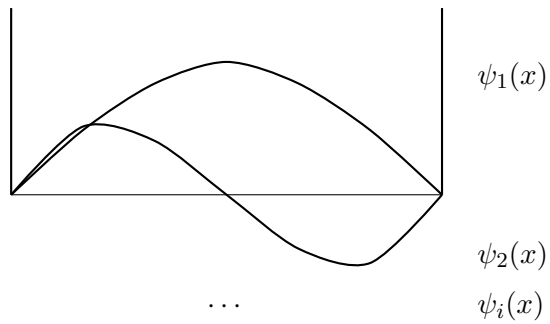


Figure 7.1: A clean schematic of the particle-in-a-box modes drawn in the spirit of the lecture board: the first few sine-like eigenfunctions, organized by node count, with labels matching Susskind's notation.

7.2 Single-Particle Quantum Mechanics Recalled

The starting point is ordinary one-particle quantum mechanics. If $|\psi\rangle$ is the state of a single particle and $|x\rangle$ denotes a state localized at position x, then the wavefunction is

$$\psi(x) = \langle x|\psi\rangle. \tag{7.1}$$

This is a probability amplitude. The corresponding probability density is

$$|\psi(x)|^2 = \psi^*(x)\psi(x), \tag{7.2}$$

and for a normalized one-particle state,

$$\int dx\, \psi^*(x)\psi(x) = 1. \tag{7.3}$$

Susskind is careful here about notation. Lower-case ψ refers to an ordinary one-particle wavefunction. It is not yet a field, not yet an operator, and not yet an object living in the many-particle Hilbert space. That distinction is one of the recurring lessons of the lecture.

There is also a useful small observation at this stage. In the one-particle problem, the integral of $|\psi(x)|^2$ is the total probability, hence 1. Since there is exactly one particle in the problem, it also happens, trivially, to be the total number of particles. Later the same-looking integral will reappear with a nontrivial meaning.

7.3 Modes, Orthogonality, and the Delta Function

Next Susskind chooses a convenient basis of one-particle states. There are many choices: the position eigenstates themselves form a basis, as do the eigenstates of any Hermitian operator. For the later construction, energy eigenstates are especially natural.

His concrete example is a particle in a box. The horizontal axis represents position, the vertical axis the potential energy, and the potential rises steeply at the walls. The first few energy eigenfunctions are essentially sine waves, ordered by their nodes.

If the one-particle eigenfunctions are written $\psi_i(x)$, then orthonormality means

$$\int dx\, \psi_i^*(x)\psi_j(x) = \delta_{ij}. \tag{7.4}$$

Completeness means that the corresponding one-particle states satisfy

$$\sum_i |i\rangle\langle i| = \mathbf{1}. \tag{7.5}$$

This is the point at which Susskind inserts what he calls a little trick, one that recurs all through the subject. Sandwich the identity operator between position states:

$$\begin{aligned} \langle x|y\rangle &= \langle x|\mathbf{1}|y\rangle && (7.6)\\ &= \sum_i \langle x|i\rangle\langle i|y\rangle && (7.7)\\ &= \sum_i \psi_i(x)\psi_i^*(y). && (7.8) \end{aligned}$$

Since $\langle x|y\rangle = \delta(x-y)$, one obtains the position-space completeness relation

$$\sum_i \psi_i(x)\psi_i^*(y) = \delta(x-y). \tag{7.9}$$

The placement of the complex conjugation depends on the bra-ket order, but this is the convention used here and it will be kept fixed throughout.

This identity is the last major one-particle ingredient needed later. In Susskind's phrase, this is the basic mathematics.

7.4 From Occupation Numbers to the Field Operator

Only after the one-particle scaffold is in place does the lecture return to the many-particle problem. The Hilbert space now changes. Instead of one-particle wavefunctions, one uses occupation-number states

$$|n_1, n_2, \ldots\rangle, \tag{7.10}$$

where n_i is the number of particles occupying the one-particle mode i. These states belong to Fock space; they do not lie in the same vector space as the one-particle functions $\psi_i(x)$.

To change occupation numbers one introduces, for each mode i, a creation operator and an annihilation operator,

$$a_i^\dagger, \qquad a_i. \tag{7.11}$$

The operator $a_i^\dagger$ creates a particle in mode i, while a_i annihilates one. The associated number operator is

$$N_i = a_i^\dagger a_i. \tag{7.12}$$

The vacuum is the state with all occupation numbers zero,

$$|0\rangle = |0, 0, \ldots\rangle. \tag{7.13}$$

A one-particle state occupying the mode i is therefore

$$|i\rangle = a_i^\dagger |0\rangle. \tag{7.14}$$

At first this oscillator language is only bookkeeping. Susskind says so explicitly. It will become more than bookkeeping later, but the immediate purpose is simply to create a formalism in which occupation numbers can be raised and lowered systematically.

Now comes the key definition:

$$\Psi(x) = \sum_i a_i\, \psi_i(x), \qquad \Psi^\dagger(x) = \sum_i a_i^\dagger\, \psi_i^*(x). \tag{7.15}$$

These are operator-valued functions of position: at each point x there is an operator. The notation is deliberately similar to the one-particle wavefunction, but the object is completely different. $\Psi(x)$ is not a one-particle state. It is an operator acting on Fock space.

The analogy with Fourier expansion is direct. If the basis functions $\psi_i(x)$ were plane waves, then the coefficients of the expansion would be the Fourier coefficients. The only novelty is that here the coefficients are not ordinary numbers but operators.

This is exactly where Susskind distinguishes c-numbers from q-numbers. The wavefunctions $\psi_i(x)$ are ordinary complex numbers as functions of x; they commute with everything. The a_i and $a_i^\dagger$ are operators. Because the wavefunctions are c-numbers, one may move them freely through the expressions without any operator-ordering issue.

Neither $\Psi(x)$ nor $\Psi^\dagger(x)$ is Hermitian. Their adjoints are each other. But the combinations

$$\Psi(x) + \Psi^\dagger(x), \qquad i\big(\Psi^\dagger(x) - \Psi(x)\big) \tag{7.16}$$

are Hermitian, hence observable. Susskind mentions this not to begin a measurement theory of Ψ immediately, but to show that field-like observables do emerge from these operator-valued functions.

The most useful way to motivate the definition is to ask what it does. Insert the resolution of the identity into the position state:

$$\begin{aligned} |x\rangle &= \mathbf{1}|x\rangle && (7.17)\\ &= \sum_i |i\rangle\langle i|x\rangle && (7.18)\\ &= \sum_i a_i^\dagger |0\rangle\, \psi_i^*(x) && (7.19)\\ &= \Psi^\dagger(x)|0\rangle. && (7.20) \end{aligned}$$

Thus $\Psi^\dagger(x)$ creates, out of the vacuum, a particle localized at the point x. This is the intuitive content of the field operator. The operators $a_i^\dagger$ create particles in definite modes; the operator $\Psi^\dagger(x)$ creates a particle at a definite position.

7.5 Playing with the Field: Number and Density

At this point Susskind does not yet write a field equation. Instead he plays with the field and asks what simple observables can be built from it. The first one is

$$\int dx\, \Psi^\dagger(x)\Psi(x). \tag{7.21}$$

This resembles the one-particle normalization integral, but the resemblance is only formal. Here Ψ and $\Psi^\dagger$ are operators.

Substituting the definitions gives

$$\int dx\, \Psi^\dagger(x)\Psi(x) = \int dx \left(\sum_i a_i^\dagger \psi_i^*(x)\right)\left(\sum_j a_j \psi_j(x)\right) \tag{7.22}$$

$$= \int dx \sum_{i,j} a_i^\dagger a_j\, \psi_i^*(x)\psi_j(x) \tag{7.23}$$

$$= \sum_{i,j} a_i^\dagger a_j \int dx\, \psi_i^*(x)\psi_j(x) \tag{7.24}$$

$$= \sum_{i,j} a_i^\dagger a_j\, \delta_{ij} \tag{7.25}$$

$$= \sum_i a_i^\dagger a_i. \tag{7.26}$$

Therefore

$$N = \int dx\, \Psi^\dagger(x)\Psi(x) = \sum_i a_i^\dagger a_i = \sum_i N_i \tag{7.27}$$

is the total-number operator.

This is the first major payoff of the field construction. The same formal structure that, in one-particle quantum mechanics, integrated to total probability, now integrates to the total number of particles in the many-particle system.

It is then natural to interpret the integrand itself as a local density:

$$n(x) = \Psi^\dagger(x)\Psi(x). \tag{7.28}$$

If one integrates this over a small spatial region R, one obtains

$$N_R = \int_R dx\, n(x), \tag{7.29}$$

the operator that counts how many particles lie in that region.

Here Susskind slows down again, because the temptation is strong to call $n(x)$ a probability density. He does not want to make that identification too quickly. In a many-particle state, what one measures in a small region is the number of particles in that region. If there are many particles, dividing by the total number often gives something that tracks a probability distribution extremely well in practice, but it is not *identically* the one-particle Born-rule density. The many-particle operator $n(x)$ is an observable, with fluctuations from state to state and measurement to measurement. It is safest to call it what it is: the particle-density operator.

7.6 A Slightly Different Thing: Energy

Having stabilized the particle-counting story, Susskind says that he wants to do a slightly different thing. The new target is the total energy.

He first restricts the discussion to independent particles, meaning particles that do not interact with one another. This is an approximation, but often a good one for dilute systems. In that

approximation the total energy is the sum over modes of occupation number times one-particle energy:

$$H = \sum_i \omega_i a_i^\dagger a_i. \tag{7.30}$$

Here ω_i is simply the energy of the ith one-particle mode. Susskind keeps the symbol ω_i from the oscillator language, while also saying explicitly that it is the energy.

To understand these ω_i, he returns to the one-particle time-independent Schrödinger equation. From here on we set $\hbar = 1$. On the board he writes

$$\left[\frac{P^2}{2m} + V(x)\right]\psi_i(x) = \omega_i \psi_i(x), \qquad P = -i\,\frac{\partial}{\partial x}. \tag{7.31}$$

For cleaner notation in the rest of the chapter, let us write the one-particle momentum operator as

$$p = -i\,\partial_x, \tag{7.32}$$

and reserve the symbol P for the total momentum operator of the field. Then the one-particle equation is

$$\left[\frac{p^2}{2m} + V(x)\right]\psi_i(x) = \omega_i \psi_i(x). \tag{7.33}$$

In three-dimensional language this becomes

$$\left[-\frac{\nabla^2}{2m} + V(x)\right]\psi_i(x) = \omega_i \psi_i(x). \tag{7.34}$$

Susskind does not solve this equation in the lecture. He uses it instead as an identity that can be inserted into the many-particle calculation. Before making the jump, he recalls one more familiar one-particle formula:

$$\int dx\, \psi^*(x)\left(-\frac{\nabla^2}{2m} + V(x)\right)\psi(x), \tag{7.35}$$

which is the expectation value of the energy in the one-particle state ψ.

Now comes the same leap that worked for particle number. Replace the little ψ by the big Ψ. The guess is that the Hamiltonian operator can be written as

$$H = \int dx\, \Psi^\dagger(x)\left(-\frac{\nabla^2}{2m} + V(x)\right)\Psi(x). \tag{7.36}$$

This is exactly the form suggested by the lecture board. Susskind writes the answer first and then proves that it is equal to the mode sum. He also continues to write $\int dx$ even after promoting the kinetic term to $-\nabla^2/2m$; the important point here is the local operator density, not the measure notation.

Expand the field operators:

$$H = \int dx \left(\sum_i a_i^\dagger \psi_i^*(x)\right)\left(-\frac{\nabla^2}{2m} + V(x)\right)\left(\sum_j a_j \psi_j(x)\right) \tag{7.37}$$

$$= \sum_{i,j} a_i^\dagger a_j \int dx\, \psi_i^*(x)\left(-\frac{\nabla^2}{2m} + V(x)\right)\psi_j(x). \tag{7.38}$$

At this point the one-particle Schrödinger equation is used:

$$\left(-\frac{\nabla^2}{2m} + V(x)\right)\psi_j(x) = \omega_j \psi_j(x). \tag{7.39}$$

Therefore

$$H = \sum_{i,j} a_i^\dagger a_j \int dx\, \psi_i^*(x)\, \omega_j \psi_j(x) \tag{7.40}$$

$$= \sum_{i,j} a_i^\dagger a_j\, \omega_j \int dx\, \psi_i^*(x) \psi_j(x) \tag{7.41}$$

$$= \sum_{i,j} a_i^\dagger a_j\, \omega_j\, \delta_{ij} \tag{7.42}$$

$$= \sum_j \omega_j a_j^\dagger a_j, \tag{7.43}$$

which is exactly the original expression for the total energy.

This calculation also clarifies an operator-ordering point that comes up in the lecture discussion. Nothing here required commuting an annihilation operator past a creation operator. The wavefunctions $\psi_i(x)$ are c-numbers, so they may be moved freely. The derivatives act on the ordinary wavefunctions, not on the operators a_i and $a_i^\dagger$.

Susskind writes this object on the board as the energy, but at the same time insists that it is an operator. In modern language one would simply say that it is the Hamiltonian operator, hence also the total-energy operator:

$$H = \int dx\, \mathcal{H}(x), \qquad \mathcal{H}(x) = \Psi^\dagger(x)\left(-\frac{\nabla^2}{2m} + V(x)\right)\Psi(x). \tag{7.44}$$

The integrand is naturally called the energy density. Susskind adds one cautionary sentence: this is a definition of the local energy density for the present Hamiltonian, not a mathematically unique notion in every context.

7.7 Why This Is a Field and What Comes Next

At this point the audience asks the natural question: why should this be called a field at all? Susskind's answer is direct. It is a quantity that depends on position, and suitable Hermitian combinations of it are observable. One can speak of it at one point of space and at another. It produces a local particle density and a local energy density. In that mathematical and physical sense, it is a field.

It is not, however, the electromagnetic field. The lecture keeps the discussion nonrelativistic. As long as the particles are slow enough that the one-particle energy is of the form $p^2/2m$ plus a potential, this is the field whose quanta are those particles. Protons, neutrons, or electrons moving slowly would all fit this pattern. The particles are the quanta of the field in exactly the same structural sense that photons are the quanta of the electromagnetic field.

The formalism also contains ordinary one-particle quantum mechanics as a special case. If one restricts to the sector in which the total number operator has eigenvalue 1, the whole construction

collapses back to the usual one-particle theory. In that sense nothing has been abandoned; the Hilbert space has simply been enlarged.

At the opposite extreme, states with very large occupation number begin to behave classically. Susskind emphasizes that expectation values then follow the same sort of equations of motion as the corresponding classical fields. For the present nonrelativistic system, the expectation value of the field operator satisfies, in essence, the time-dependent Schrödinger equation:

$$i\,\partial_t \langle \Psi(x,t)\rangle = \left(-\frac{\nabla^2}{2m} + V(x)\right)\langle \Psi(x,t)\rangle. \tag{7.45}$$

This statement is asserted rather than derived in detail in the lecture, but it is the crucial link between the quantum field and its classical limit.

A late aside in the lecture looks ahead to relativity. There Susskind warns that particle position is not the best concept, and therefore particle density is not generally the right observable. Energy density is much better behaved. He mentions, only heuristically, that for electromagnetic radiation quantities such as $E^2 + B^2$ play the role of energy density. This aside is useful as intuition, but it lies outside the main nonrelativistic derivation.

The final step of the lecture is another preview. For a one-particle wavefunction,

$$\langle p\rangle_\psi = \int dx\, \psi^*(x)\,(-i\,\partial_x)\,\psi(x). \tag{7.46}$$

By the same replacement rule that carried probability density to number density, and one-particle energy to field Hamiltonian, one is led to the total momentum operator of the field:

$$P = \int dx\, \Psi^\dagger(x)\,(-i\,\partial_x)\,\Psi(x). \tag{7.47}$$

Susskind does not fully derive this in the lecture. He presents it as the next natural observable and a teaser for the next meeting, stressing that it is an operator, not an expectation value. An operator and a state together produce an expectation value; the operator alone is the observable itself.

7.8 Summary

The lecture reconstructs a simple quantum field from the particle side. The motivation is the need to describe arbitrary and changing particle number. The construction begins with ordinary one-particle wavefunctions, orthonormal modes, and the completeness relation, then passes to Fock space, where occupation numbers are changed by creation and annihilation operators. Out of these ingredients one defines the field operators $\Psi(x)$ and $\Psi^\dagger(x)$, interprets $\Psi^\dagger(x)$ as creating a particle at the point x, derives the total-number operator and the particle-density operator, and finally rewrites the mode-sum energy as the Hamiltonian

$$H = \int dx\, \Psi^\dagger(x)\left(-\frac{\nabla^2}{2m} + V(x)\right)\Psi(x). \tag{7.48}$$

That is the lecture's central payoff: the many-particle system can be described as a field whose quanta are the particles themselves. The next step, deferred to the following lecture, is to use these field operators to represent actual processes such as scattering and decay.

Chapter 8

Mixing, Oscillation, Electric Dipole Moments, and Field Operators

This chapter follows Leonard Susskind's *Advanced Quantum Mechanics* Lecture 8 and is curated by LazyingArt LLC. The lecture begins from a live question about neutrinos, uses that question to build a clean two-state model of mixing, pauses for a long interlude on what an electric dipole moment does and does not tell us about the electron, and then explicitly returns to second quantization. The central theme is simple but powerful: the states in which systems are prepared need not be the states that diagonalize the Hamiltonian. Once that distinction is kept in view, the rest of the lecture unfolds naturally.

8.1 Mixing as the Real Issue

A student asks whether neutrinos, if they have mass, must all have the same mass; if not, how can they mix without violating energy conservation? Susskind's answer is immediate and conceptual. Mass is energy, and the energy eigenstates do not mix with each other. What mixes are the physically named states, the states of *type*. In a schematic notation,

$$|\text{type}\rangle = \sum_n c_n |E_n\rangle. \tag{8.1}$$

Thus the right statement is not that energy eigenstates change into one another, but that the states called "electron neutrino" or "muon neutrino" need not themselves be energy eigenstates.

He emphasizes that this is a common pattern in quantum mechanics and then says, in effect, that a simple example is enough to see it. The abstract neutrino question is therefore translated into a very concrete one-dimensional problem.

8.2 Symmetric Double Well and Approximate Localized States

Consider a particle moving in a symmetric double-well potential with a high barrier between the two minima. If the barrier were taken to be infinitely high, then the left and right wells would decouple completely. One could solve the left-hand well as though the right-hand well did not exist,

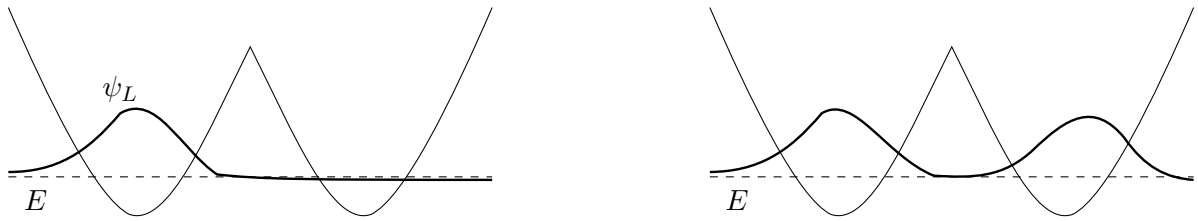


Figure 8.1: A clean redraw of the opening board sketches. On the left is the one-sided trial state ψ_L, obtained by pretending the barrier is effectively infinite. On the right the same wavefunction has been continued symmetrically through the barrier, illustrating the small but nonzero leakage that eventually produces the true even ground state.

and likewise for the right-hand well. The approximate ground-state wavefunctions would then be a left-localized state ψ_L and a right-localized state ψ_R, both with the same energy E.

In each isolated well the ground state has the familiar qualitative properties: it is smooth, has no nodes, and sits at the lowest possible energy consistent with confinement. Susskind explicitly asks the room to imagine the barrier as a brick wall. It is only a good approximation, but it is a useful one.

Quantum mechanically, however, the barrier is not truly infinite. The wavefunction in the barrier region is not exactly zero; it is only very small. This tiny leakage is the whole story. It is what converts the left-right approximation into a genuine mixing problem.

8.3 Parity Eigenstates and Exponential Level Splitting

At this point Susskind interrupts the naive picture with a theorem: if the potential is symmetric, then the exact energy eigenfunctions can be chosen to be either symmetric or antisymmetric under reflection.

Theorem 8.1. *If $V(x) = V(-x)$ in one dimension, then the eigenfunctions of the Hamiltonian may be chosen to satisfy either*

$$\psi(x) = \psi(-x) \qquad \text{or} \qquad \psi(x) = -\psi(-x).$$

Now comes the correction. The one-sided states ψ_L and ψ_R are neither symmetric nor antisymmetric, so they cannot be exact eigenstates of the full Hamiltonian. They are only good approximate states when the barrier is high. The true eigenstates are the symmetric and antisymmetric combinations,

$$\psi_+ = \frac{\psi_L + \psi_R}{\sqrt{2}}, \tag{8.2}$$

$$\psi_- = \frac{\psi_L - \psi_R}{\sqrt{2}}. \tag{8.3}$$

This is exactly how the blackboard notation settles. Susskind then labels the energies by

$$H\psi_+ = (E_1 - \epsilon)\psi_+, \tag{8.4}$$

$$H\psi_- = (E_1 + \epsilon)\psi_-. \tag{8.5}$$

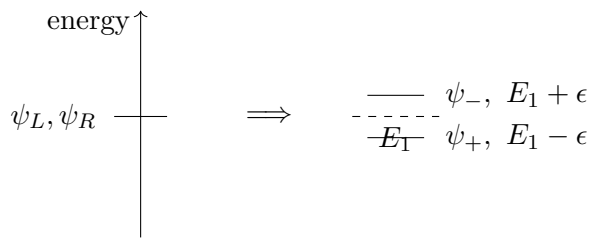


Figure 8.2: The board logic of the tunnel splitting. The approximate left and right states are nearly degenerate, but the exact symmetric and antisymmetric eigenstates are split by a small amount 2ϵ.

The total splitting is therefore

$$\Delta E = (E_1 + \epsilon) - (E_1 - \epsilon) = 2\epsilon. \tag{8.6}$$

The ordering of the levels is part of the lecture's physical point. The symmetric state is slightly lower in energy because it is smoother through the barrier region; allowing the wavefunction to leak and continue symmetrically lowers the energy by a tiny amount. The antisymmetric state is slightly higher because the first excited state in one dimension has one node, and in this symmetric problem that node sits at the midpoint. Extra curvature costs kinetic energy.

Susskind stresses that the effect is very small, exponentially small in the barrier width and height. He does not need an explicit WKB formula here; the important point is that the splitting is real, but tiny. He also remarks that the overall sign of a wavefunction carries no physical information. Replacing ψ_- by $-\psi_-$ does not change the state.

8.4 Time Evolution and Left-Right Oscillation

Only after the static picture is clear does Susskind ask the dynamical question: what happens if we start with a particle placed in the left-hand well? This is the decisive step, because oscillation is the time-dependent face of the same mixing phenomenon.

The inverse relations are

$$\psi_L = \frac{\psi_+ + \psi_-}{\sqrt{2}}, \tag{8.7}$$

$$\psi_R = \frac{\psi_+ - \psi_-}{\sqrt{2}}. \tag{8.8}$$

So an initially left-localized state is

$$\psi(0) = \psi_L = \frac{\psi_+ + \psi_-}{\sqrt{2}}. \tag{8.9}$$

It is not an energy eigenstate, and therefore it must evolve nontrivially.

Remark 8.2. In the lecture the signs in the time-dependent phases briefly wobble, and Susskind explicitly catches the slip. The written notes use the standard Schrödinger convention e^{-iEt}. The only physically important quantity here is the relative phase between the two energy eigenstates.

With that convention,

$$\psi(t) = \frac{1}{\sqrt{2}} \left(e^{-i(E_1-\epsilon)t}\psi_+ + e^{-i(E_1+\epsilon)t}\psi_- \right). \tag{8.10}$$

This is the cleaned version of the board formula. Now factor out the common phase:

$$\psi(t) = \frac{e^{-iE_1 t}}{\sqrt{2}} \left(e^{i\epsilon t}\psi_+ + e^{-i\epsilon t}\psi_- \right). \tag{8.11}$$

At this point the lecture turns from bookkeeping to geometry: one phase runs one way around the unit circle, the other runs the opposite way. The state changes only because these two phases drift out of step.

To see the left-right oscillation explicitly, substitute the definitions of $\psi_\pm$:

$$\psi(t) = \frac{e^{-iE_1 t}}{2} \left[e^{i\epsilon t}(\psi_L + \psi_R) + e^{-i\epsilon t}(\psi_L - \psi_R) \right] \tag{8.12}$$

$$= \frac{e^{-iE_1 t}}{2} \left[(e^{i\epsilon t} + e^{-i\epsilon t})\psi_L + (e^{i\epsilon t} - e^{-i\epsilon t})\psi_R \right] \tag{8.13}$$

$$= e^{-iE_1 t} \left[\psi_L \cos(\epsilon t) + i\, \psi_R \sin(\epsilon t) \right]. \tag{8.14}$$

This is the key worked derivation of the chapter. In the two-state approximation, with ψ_L and ψ_R orthonormal, the probabilities are

$$P_L(t) = \cos^2(\epsilon t), \qquad P_R(t) = \sin^2(\epsilon t). \tag{8.15}$$

So the system oscillates. At intermediate times it is genuinely in superposition; for example, when $\epsilon t = \pi/4$, the probabilities on the two sides are equal. The lecture insists on this continuity: the phases evolve continuously, and the localization changes continuously with them.

The blackboard derivation isolates the first time at which the relative sign flips. One asks when

$$e^{-i\epsilon t} = -\, e^{i\epsilon t}, \tag{8.16}$$

which is the visible milestone on the board. Multiplying through gives

$$e^{2i\epsilon t} = -1. \tag{8.17}$$

For the first occurrence,

$$2\epsilon t = \pi, \qquad \text{so} \qquad t_{L\to R} = \frac{\pi}{2\epsilon}. \tag{8.18}$$

At that instant,

$$\psi(t_{L\to R}) \propto \frac{\psi_+ - \psi_-}{\sqrt{2}} = \psi_R. \tag{8.19}$$

The overall phase is irrelevant. What matters is that the particle initially placed on the left has evolved into the right-localized state. After another equal interval it swings back again. That is why Susskind says that mixing and oscillation go together.

There is one spoken slip in the lecture immediately after the correct formula: Susskind first says that if ϵ is small, the transfer time can be very long, and then momentarily says the opposite. The equation is unambiguous. Small ϵ means slow oscillation and a long transfer time.

8.5 Neutrino, Ammonia, and Spin Analogies

With the double well understood, Susskind returns to the original neutrino question. The left and right wells now become a template. In a two-state caricature,

$$|\nu_e\rangle \sim \frac{|m_1\rangle + |m_2\rangle}{\sqrt{2}}, \qquad |\nu_\mu\rangle \sim \frac{|m_1\rangle - |m_2\rangle}{\sqrt{2}}, \tag{8.20}$$

where the mass eigenstates play the role of ψ_+ and ψ_-, and the flavor states play the role of ψ_L and ψ_R. An electron neutrino is produced in processes involving an electron, just as a particle can be "plunked" into the left well. The Hamiltonian, however, evolves the mass eigenstates. As the relative phase accumulates, the same neutrino can later behave as a muon neutrino.

That is the lecture's answer to the apparent conflict with energy conservation: the energy eigenstates do exactly what energy eigenstates are supposed to do. It is the non-energy-eigenstate flavor state that evolves. This is also why oscillation is evidence for neutrino mass. If the neutrino were truly massless, there would be no such mass splitting and no such oscillation phenomenon.

Susskind also discusses ammonia inversion. There the nitrogen atom can sit on one side or the other of the plane of hydrogen atoms, and tunneling through that plane produces the same symmetric and antisymmetric combinations. In ammonia the coupling really is tunneling through a barrier. In the neutrino problem, by contrast, there is no literal barrier in space. There is instead a coupling in the Hamiltonian that mixes electron and muon neutrino states. The mathematics is the same even though the microscopic origin of the coupling is different.

He then says, in effect, that we have already seen this same algebra elsewhere: in spin. Take the two spin states $|\uparrow_z\rangle$ and $|\downarrow_z\rangle$. With no field they are degenerate. Now turn on a weak magnetic field in the x-direction. The energy eigenstates are no longer the z-eigenstates, but the x-polarized states. Following the lecture's temporary language,

$$|L_x\rangle = \frac{|\uparrow_z\rangle + |\downarrow_z\rangle}{\sqrt{2}}, \tag{8.21}$$

$$|R_x\rangle = \frac{|\uparrow_z\rangle - |\downarrow_z\rangle}{\sqrt{2}}. \tag{8.22}$$

These are the direct analogues of ψ_+ and ψ_-. If their energies are $E \mp \epsilon$, then an initially up spin evolves as

$$|\uparrow_z, t\rangle = e^{-iEt}\left[\cos(\epsilon t)\,|\uparrow_z\rangle + i\sin(\epsilon t)\,|\downarrow_z\rangle\right]. \tag{8.23}$$

This is the same formula as for the double well, only now the oscillation is observed as spin precession. After one half period the spin is certainly down; after another half period it is back up. The weakness of the field matters only because it makes the oscillation slow enough to compare with the neutrino story. The mechanism itself does not depend on tunneling. It depends only on off-diagonal mixing in the Hamiltonian.

The lecture briefly notes that real neutrinos involve more than two states: the tau neutrino also mixes in. That makes the algebra more elaborate, but not conceptually different.

8.6 Interlude: Electric Dipole Moment, Symmetry, and the "Electron Sphere" Claim

At this point Susskind deliberately changes register. Since the lecture has already drifted into real particle-physics questions, he pauses the formal development and responds to a popular article claiming that a null electron electric-dipole-moment measurement shows that "the electron is a sphere." His view is that this is badly misleading.

8.6.1 What was measured?

What was measured, or rather bounded, is the electric dipole moment of the electron. For separated charges $\pm q$, the dipole moment is

$$\mathbf{d} = q\,\boldsymbol{\ell}, \tag{8.24}$$

with $\boldsymbol{\ell}$ the separation vector. A nonzero dipole moment means that the charge distribution is displaced relative to some center.

For the electron, however, the relevant question is not merely whether the charge is off-center in some arbitrary direction. The electron has spin and therefore an axis. It also has a magnetic moment. The physically meaningful question is whether there is a correlation between the displacement of charge and that axis:

$$\mathbf{d}_e \parallel \boldsymbol{\mu}_e \qquad \text{or equivalently} \qquad \mathbf{d}_e \parallel \mathbf{S}. \tag{8.25}$$

That is what Susskind means by the electron's electric dipole moment. Conceptually, one asks whether placing the spin in an electric field would split the two orientations:

$$\Delta E_{\text{EDM}} = -\mathbf{d}_e \cdot \mathbf{E}. \tag{8.26}$$

The actual tabletop experiment is more subtle than this elementary picture, but this is the basic idea.

8.6.2 Why no EDM does not mean sphere

The first correction is elementary. A body can have no dipole moment and still not be spherical. An ellipsoid, for example, can have zero dipole moment and yet carry a quadrupole moment. One can also imagine higher multipoles with no dipole and no quadrupole. So the statement "no EDM" does not by itself even come close to proving "sphere."

Susskind then turns the argument into a quantum-mechanical one. Imagine a lopsided dumbbell-like molecule. Classically it certainly does not look spherical. But if its ground state has angular momentum zero, then because angular momentum generates rotations,

$$J = 0 \qquad \Longrightarrow \qquad \text{rotationally invariant ground state.} \tag{8.27}$$

So the quantum state can be rotationally symmetric even though the underlying object is not a classical sphere. One may think of the state as a superposition of all orientations.

This immediately raises a measurement question. A quick snapshot of orientation is itself a measurement of angle. Such a measurement forces the system into a state of definite orientation, and therefore kicks its angular momentum. Susskind presents this as the angular-momentum version

of the ordinary position-momentum uncertainty principle. Whether the lopsidedness can actually be resolved depends on the energy needed to excite rotational states. For a macroscopic dumbbell that energy spacing is tiny, so ordinary probes can resolve orientation. For a tightly bound microscopic system, the spacing can be so large that ordinary probes only see the rotationally averaged state.

The lecture pushes this point all the way to hadronic scales. A meson, thought of loosely as a two-body dumbbell, may look spherical to low-energy probes because the energy required to excite rotational structure is extremely large. "Sphere or not" therefore becomes a question about the resolving power of the experiment.

Susskind then makes a sharper point about the electron. Independently of any EDM measurement, the electron is a spin-$\frac{1}{2}$ particle. It has an axis. It has a magnetic moment. In that very direct sense it is already not a spherically symmetric scalar object. So the slogan that the electron "is a sphere" was, in his view, already wrong before one even began discussing the EDM bound.

8.6.3 Parity and time reversal

The real content of the EDM discussion is symmetry. Under parity, an electric dipole moment behaves as an ordinary polar vector, while a magnetic moment behaves as an axial vector:

$$P: \qquad \mathbf{d}_e \to -\mathbf{d}_e, \qquad\qquad \boldsymbol{\mu}_e \to \boldsymbol{\mu}_e. \tag{8.28}$$

Under time reversal, the circulating current that defines the magnetic moment reverses, but a static electric displacement does not:

$$T: \qquad \mathbf{d}_e \to \mathbf{d}_e, \qquad\qquad \boldsymbol{\mu}_e \to -\boldsymbol{\mu}_e. \tag{8.29}$$

Therefore a nonzero EDM aligned with the electron's spin axis is forbidden by exact parity symmetry and also by exact time-reversal symmetry:

$$(P \text{ exact}) \Rightarrow \mathbf{d}_e = 0, \qquad (T \text{ exact}) \Rightarrow \mathbf{d}_e = 0. \tag{8.30}$$

This is precisely the logic Susskind tries to draw geometrically on the board. Under mirror reflection, the magnetic axis is reproduced with the same orientation, whereas a displaced charge distribution would reverse its relation to that axis. If the world had exact left-right symmetry and there were only one kind of electron, the mirrored electron would have to be the same electron, and the only consistent possibility would be zero EDM. Under time reversal the current reverses, and hence the magnetic moment flips, while the displaced charge does not. The same conclusion follows again.

Now parity is not a good symmetry of the weak interactions, so parity alone does not save us. But time-reversal violation in the standard model is extremely small, and so the standard-model electron EDM is correspondingly tiny. The experiment he is discussing did not detect any EDM. That is not evidence that the electron is spherical. It is evidence that any EDM is very strongly suppressed, which is good for the standard model and constraining for possible extensions beyond it.

The lecture's punchline is therefore careful and modest: the null result says the electron does not have an observable electric dipole moment at the measured level. It does not say the electron is a sphere.

8.7 Return to Second Quantization: Fourier Transforms, Momentum Modes, and Locality

After the break Susskind explicitly restarts the main thread: "we were doing second quantization." He does so gently, beginning with ordinary Fourier transforms in single-particle quantum mechanics and then lifting the same structure to field operators.

8.7.1 From $\psi(x)$ to $\tilde{\psi}(p)$

He first reminds the class that little ψ, not capital Ψ, is the wavefunction of a single particle. In position space,

$$P(x) = |\psi(x)|^2 \tag{8.31}$$

is the probability density to find the particle at position x. The momentum-space wavefunction is its Fourier transform,

$$\tilde{\psi}(p) = \int dx\, \psi(x)\, \frac{e^{-ipx}}{\sqrt{2\pi}}, \tag{8.32}$$

and the inverse relation is

$$\psi(x) = \int dp\, \tilde{\psi}(p)\, \frac{e^{ipx}}{\sqrt{2\pi}}. \tag{8.33}$$

Then

$$P(p) = |\tilde{\psi}(p)|^2 \tag{8.34}$$

is the probability density for momentum p.

For a free particle,

$$H = \frac{p^2}{2m}, \tag{8.35}$$

so momentum eigenstates are also energy eigenstates, and their position-space wavefunctions are plane waves,

$$\psi_p(x) \propto e^{ipx}. \tag{8.36}$$

This is the point of departure for the second-quantized version.

8.7.2 Field operators in position and momentum space

Susskind then goes back to the field operator. On the blackboard he writes the annihilation and creation pieces as $\Psi_-(x)$ and $\Psi_+(x)$. Keeping that notation for a moment,

$$\Psi_-(x) = \sum_i a_i\, \psi_i(x), \tag{8.37}$$

$$\Psi_+(x) = \sum_i a_i^\dagger\, \psi_i^*(x). \tag{8.38}$$

Here $\psi_i(x)$ is the ordinary one-particle wavefunction of the i-th mode, while a_i and $a_i^\dagger$ annihilate and create particles in that mode. From this point on it is convenient to rename

$$\Psi(x) \equiv \Psi_-(x), \qquad \Psi^\dagger(x) \equiv \Psi_+(x).$$

For a free particle the discrete mode label i is replaced by the continuous momentum label p. Using the plane waves e^{ipx}, one obtains the standard reconstructed formulas

$$\Psi(x) = \int \frac{dp}{\sqrt{2\pi}}\, a(p)\, e^{ipx}, \tag{8.39}$$

$$\Psi^\dagger(x) = \int \frac{dp}{\sqrt{2\pi}}\, a^\dagger(p)\, e^{-ipx}. \tag{8.40}$$

These are the field-operator analogues of the ordinary Fourier transform. They can be inverted exactly as before:

$$a(p) = \int \frac{dx}{\sqrt{2\pi}}\, \Psi(x)\, e^{-ipx}, \tag{8.41}$$

$$a^\dagger(p) = \int \frac{dx}{\sqrt{2\pi}}\, \Psi^\dagger(x)\, e^{ipx}. \tag{8.42}$$

The normalization in this part of the live lecture is not completely legible on the board, but the standard $1/\sqrt{2\pi}$ convention is the natural cleaned reconstruction.

The conceptual content is straightforward. $\Psi^\dagger(x)$ creates a particle at position x, while $a^\dagger(p)$ creates a particle with momentum p. Position-space and momentum-space creation operators are Fourier conjugates of one another, just as $\psi(x)$ and $\tilde{\psi}(p)$ are Fourier conjugates in ordinary quantum mechanics.

8.7.3 Bosonic commutators and the independence of separated measurements

The lecture then turns to commutators. For bosons,

$$[a_i, a_j^\dagger] = \delta_{ij}, \qquad [a_i, a_j] = 0, \qquad [a_i^\dagger, a_j^\dagger] = 0. \tag{8.43}$$

The continuum version is

$$[a(p), a^\dagger(p')] = \delta(p - p'). \tag{8.44}$$

Now compute the equal-time field commutator from the discrete expansion:

$$[\Psi(x), \Psi^\dagger(y)] = \sum_{i,j} \psi_i(x)\psi_j^*(y)\, [a_i, a_j^\dagger] \tag{8.45}$$

$$= \sum_i \psi_i(x)\psi_i^*(y) \tag{8.46}$$

$$= \delta(x - y), \tag{8.47}$$

where the last step uses completeness of the one-particle basis. Likewise,

$$[\Psi(x), \Psi(y)] = 0, \qquad [\Psi^\dagger(x), \Psi^\dagger(y)] = 0. \tag{8.48}$$

Susskind emphasizes that the most important point is not that Ψ is non-Hermitian. One can always take real and imaginary parts to obtain Hermitian combinations. The important point is the delta function. When $x \neq y$, the commutator vanishes. Measurements of the field at different points therefore do not interfere with one another. When $x = y$, the nontrivial commutator reappears, and the same-point real and imaginary parts cannot be measured simultaneously without mutual disturbance.

This is the lecture's locality heuristic in bosonic second quantization. Commuting operators at separated points express the independence of separated measurements. Susskind presses this toward a relativistic intuition: if measurements at spacelike separated points failed to commute, one would seem to have a mechanism by which a measurement here gives a kick there. In the lecture he uses this to answer a question about entanglement. Correlation is not influence. The vanishing commutator at separated points is exactly what prevents the formalism from turning entanglement into superluminal causal action.

He ends by looking ahead. Fermions will not satisfy these commutators; they satisfy anticommutation relations instead. That sounds alarming if one tries to interpret the fermion field itself as a directly measurable local observable. The resolution, postponed to the next lecture, is that fermion fields are not observables in the same direct sense as bosonic fields.

8.8 Summary

The lecture begins with a neutrino puzzle but quickly reveals a much broader lesson. Energy eigenstates do not mix; the prepared states do. In the symmetric double well this distinction is visible on the board: the approximate left and right states are not exact eigenstates, the exact eigenstates are the symmetric and antisymmetric combinations, and the tiny splitting between them drives left-right oscillation.

Susskind then shows that the same algebra reappears in neutrino oscillation, ammonia inversion, and spin precession. The long EDM interlude serves a different purpose: it disentangles a symmetry statement from a careless popular slogan. A null electron EDM measurement does not prove that the electron is spherical; it shows only that any EDM is strongly suppressed. Finally, returning to second quantization, the lecture replays the familiar Fourier-transform structure at the operator level and ends with the bosonic equal-time commutator, whose main physical meaning is the independence of measurements at different points. The next lecture, by design, begins where this one stops: with fermions and anticommutators.

Chapter 9

Quantum Fields, Local Interactions, and the One-Dimensional Dirac Equation

These notes follow Leonard Susskind's ninth lecture in *Advanced Quantum Mechanics* and are curated by LazyingArt LLC as companion notes. The lecture unfolds in two large movements. It begins as a continuation of the previous discussion of quantum fields: creation and annihilation operators, the second-quantized Schrödinger Hamiltonian, and the question of how conservation laws are actually encoded in that Hamiltonian. From there Susskind introduces local interaction terms, explains what they mean physically, and shows how repeated action of the Hamiltonian generates the first diagrammatic processes. Only then does he pivot to relativity, reject the square-root Hamiltonian as an unsatisfactory starting point, and build the one-dimensional Dirac equation step by step.

9.1 Writing the Hamiltonian Again

Susskind opens by saying, in effect, that the first task is simply to write down again the Hamiltonian for a simple quantum field whose quanta satisfy the nonrelativistic Schrödinger equation. He writes dx as though the problem were one-dimensional, but explicitly notes that x may stand for all spatial coordinates. In that notation,

$$H = \int dx\, \psi^\dagger(x) \left[-\frac{\nabla^2}{2m} + V(x) \right] \psi(x). \tag{9.1}$$

The operator $\psi^\dagger(x)$ creates a particle at position x, and $\psi(x)$ annihilates one there. The combination

$$n(x) = \psi^\dagger(x)\psi(x) \tag{9.2}$$

is therefore the density of particles at the point x. The potential term is then read exactly as one would read it classically:

$$\int dx\, V(x)\psi^\dagger(x)\psi(x) = \int dx\, V(x)\, n(x). \tag{9.3}$$

It counts how many particles sit at x and assigns to each of them the potential energy $V(x)$.

At this stage the particles do not interact with one another; they are only moving in an external potential. Susskind then makes a seemingly trivial simplification that turns out to be the bridge to the next question. He takes $V(x)$ to be a constant V_0. Then

$$\int dx\, V_0\, \psi^\dagger(x)\psi(x) = V_0 \int dx\, \psi^\dagger(x)\psi(x) = V_0\, N, \tag{9.4}$$

where

$$N = \int dx\, n(x) \tag{9.5}$$

is the total particle-number operator. A constant potential exerts no force, so dynamically it does nothing interesting. But it does make the meaning of the term transparent: it simply gives every particle the same amount of energy. In particular, this is exactly where one could keep track of a per-particle rest-energy contribution such as mc^2. Susskind keeps this otherwise uninteresting constant term on the board because it isolates the number-counting structure before he asks the real question: how does one see momentum conservation directly from the Hamiltonian?

9.2 What the Hamiltonian Does

The next move is not abstract but operational. Susskind asks what it even means for momentum to be conserved. His answer is phrased in terms of time evolution: the Hamiltonian is the operator that updates the state of the system from one instant to the next.

At this point he deliberately changes notation. Since ψ has already been used for field operators and also for single-particle wavefunctions, he refuses to use it a third time for the state vector. He writes the many-body state as $|\phi(t)\rangle$. For a small time interval ϵ,

$$|\phi(t+\epsilon)\rangle = (1 - i\epsilon H)|\phi(t)\rangle. \tag{9.6}$$

Equivalently, one may say that this is the first step in the expansion of the Schrödinger evolution operator.

Momentum conservation is then rephrased in the most concrete possible way: if H acts on a state of definite total momentum, it must return a state with the same total momentum. Susskind does not yet want to invoke translation invariance as an abstract theorem. He wants to see the mechanism. That means rewriting the Hamiltonian in the momentum basis and watching the conservation law emerge from the integral over space.

9.3 How Momentum Conservation Appears

The lecture now turns into a worked Fourier-transform exercise. The annihilation operator at position x is expanded as

$$\psi(x) = \int \frac{dp}{\sqrt{2\pi}}\, \tilde{\psi}(p) e^{ipx}, \tag{9.7}$$

and the adjoint field as

$$\psi^\dagger(x) = \int \frac{dq}{\sqrt{2\pi}}\, \tilde{\psi}^\dagger(q) e^{-iqx}. \tag{9.8}$$

Here $\tilde{\psi}(p)$ annihilates a particle of momentum p, while $\tilde{\psi}^\dagger(q)$ creates one of momentum q. Susskind is casual about 2π-bookkeeping in speech; with the convention used here,

$$\int dx\, e^{i(p-q)x} = 2\pi\, \delta(p-q). \tag{9.9}$$

Substituting these into the number term gives

$$\int dx\, \psi^\dagger(x)\psi(x) = \int \frac{dp\, dq}{2\pi}\, \tilde{\psi}^\dagger(q)\tilde{\psi}(p) \int dx\, e^{i(p-q)x} \tag{9.10}$$

$$= \int dp\, dq\, \tilde{\psi}^\dagger(q)\tilde{\psi}(p)\, \delta(p-q) \tag{9.11}$$

$$= \int dp\, \tilde{\psi}^\dagger(p)\tilde{\psi}(p). \tag{9.12}$$

This is the cleanest possible illustration of the rule. The operator annihilates a particle of momentum p and then recreates a particle of the same momentum. The intermediate expression contains a delta function because the spatial integral forces $p = q$. In the lecture Susskind pauses over the standard mnemonic: when $p \neq q$, the phase oscillates and integrates to zero; when $p = q$, the integral is singular. The precise object interpolating between those two statements is the Dirac delta function.

He then immediately generalizes. Suppose a Hamiltonian term is a local product of many fields, possibly of several species, but still integrated over a single spatial variable x. Using his lecture convention that the p's go with annihilation operators and the q's with creation operators, one finds schematically

$$\int dx \prod_a \psi^\dagger_{i_a}(x) \prod_b \psi_{j_b}(x) \quad \longrightarrow \quad \delta\left(\sum_b p_b - \sum_a q_a\right). \tag{9.13}$$

So the integral over space does *not* say that each incoming momentum must equal a corresponding outgoing momentum. It says only that the total incoming momentum equals the total outgoing momentum. This is exactly the form of momentum conservation needed for genuine interactions. Susskind remarks that the deeper reason is translation invariance, but he wants the audience to see the machinery explicitly.

He then returns to the kinetic term so that the same machinery can be seen once more in a slightly less trivial setting:

$$\int dx\, \psi^\dagger(x) \left(-\frac{\partial_x^2}{2m}\right) \psi(x) = \int \frac{dp\, dq}{2\pi}\, \tilde{\psi}^\dagger(q)\tilde{\psi}(p)\, \frac{p^2}{2m} \int dx\, e^{i(p-q)x} \tag{9.14}$$

$$= \int dp\, \frac{p^2}{2m}\, \tilde{\psi}^\dagger(p)\tilde{\psi}(p). \tag{9.15}$$

The same delta function again enforces equality of momentum in and momentum out. The only difference is that each mode is now weighted by $p^2/2m$, exactly as required for the kinetic energy.

There is one further point Susskind pauses to emphasize. The kinetic term appears with a minus sign, but kinetic energy should be positive. By integrating by parts once and dropping boundary terms,

$$\int dx\, \psi^\dagger(x)(-\partial_x^2)\psi(x) = \int dx\, (\partial_x\psi^\dagger(x))(\partial_x\psi(x)). \tag{9.16}$$

In this form the positivity is manifest: one sees the spatial derivative of the field squared, just as in other field theories.

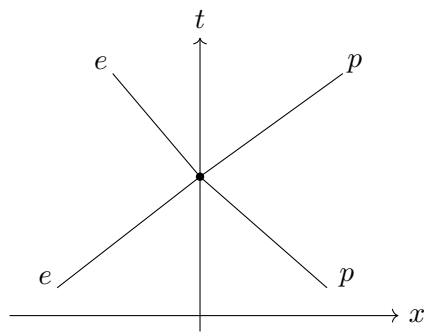


Figure 9.1: A local contact interaction: the operator acts only when the two worldlines meet at the same spacetime point.

9.4 Local Interaction Terms

With the formal mechanism in place, Susskind asks the next natural question: why would one ever write a Hamiltonian containing more than two fields at the same point? His answer is pragmatic. If experiment tells you that particles can undergo a certain local process, then the Hamiltonian, being the operator that updates the state, must contain something that produces that process.

His first example is a toy short-range interaction between two bosonic species that he informally calls the electron and the proton. The realistic electron-proton problem would require fermion operators and the Coulomb interaction, but those complications are deliberately set aside. The point is the local operator

$$H_{\text{contact}} = g \int dx\, \psi_e^\dagger(x)\psi_p^\dagger(x)\psi_e(x)\psi_p(x), \tag{9.17}$$

to be understood together with the separate kinetic terms of the two species. This interaction acts only when the electron and proton are at the same point x. If they are not, the operator gives zero. If they are, it annihilates both and recreates both at that same point. That is why Susskind interprets it as a short-range scattering operator.

This is also the first place where the earlier momentum argument pays off. The total momentum is conserved because the interaction is integrated over space, but the individual momenta of the electron and proton are not separately conserved.

The next example is not scattering but decay. Suppose experiment shows that a particle A can disappear and produce two particles B and C at the same point. Then one writes

$$H_{A\to B+C} = g \int dx\, \psi_B^\dagger(x)\psi_C^\dagger(x)\psi_A(x). \tag{9.18}$$

Because all points of space are to be treated equally, the interaction is integrated over x. Once again, that integral guarantees momentum conservation:

$$p_A = p_B + p_C \tag{9.19}$$

at the level of total incoming and outgoing momentum.

But this is not yet acceptable as part of a Hamiltonian, because a Hamiltonian must be hermitian:

$$H = H^\dagger. \tag{9.20}$$

Susskind emphasizes that this is equivalent to unitary time evolution and hence to conservation of probability. The hermitian conjugate of the decay term is

$$g \int dx\, \psi_A^\dagger(x)\psi_C(x)\psi_B(x), \tag{9.21}$$

so the full interaction is

$$H_{\text{int}} = g \int dx\, \psi_B^\dagger(x)\psi_C^\dagger(x)\psi_A(x) + g \int dx\, \psi_A^\dagger(x)\psi_C(x)\psi_B(x). \tag{9.22}$$

In this bosonic toy setting the different species commute, so reversing the order under hermitian conjugation causes no practical difficulty. The important physical statement is that once the Hamiltonian allows $A \to B + C$, it must also allow $B + C \to A$.

The constant g is the coupling constant. It measures the interaction strength. If g is small, even particles that meet at the same point interact only with small probability; if g is large, they interact more readily. Susskind stresses that these coupling constants are, in general, experimental input, though symmetry principles can sometimes relate one coupling to another. He even remarks that a four-field term of the same general kind would model processes such as beta decay.

9.5 Going One Step Further: H^2 and Diagrammatics

Having described what a single insertion of the Hamiltonian can do, Susskind goes one step further. The first-order update rule is only the start of the exponential:

$$|\phi(t+\epsilon)\rangle = e^{-i\epsilon H}|\phi(t)\rangle = \left(1 - i\epsilon H - \frac{\epsilon^2}{2}H^2 + \frac{i\epsilon^3}{3!}H^3 + \cdots\right)|\phi(t)\rangle. \tag{9.23}$$

The second-order term is the new ingredient. It means that the Hamiltonian acts twice, and therefore that two local interactions can occur in sequence within the perturbative expansion.

For the interaction $A \leftrightarrow B + C$, one insertion of H can turn A into $B + C$, or $B + C$ into A. But H^2 can do more. A first insertion can take $B + C \to A$, and a second can take $A \to B + C$ again. The net result is a scattering amplitude for the B-C system. Likewise, if one starts with an A, the sequence

$$A \longrightarrow B + C \longrightarrow A \tag{9.24}$$

produces a self-energy contribution: the particle A acquires an amplitude to behave as though it were temporarily a nearby B-C pair. If another particle is present, one also gets exchange processes. The same simple Hamiltonian term generates a large family of reactions.

Each local interaction is called a vertex, and each vertex carries one factor of the coupling constant g. Thus a second-order process carries g^2. This is one of the great powers of quantum field theory in Susskind's presentation: a very small number of local terms in the Hamiltonian correlates a very large amount of scattering data.

But he is equally insistent on a warning. The pictures one draws for these processes are not little classical movies. They are shorthand for pieces of the power-series expansion of the evolution operator. One should not over-literalize them. In the lecture, a student asks about energy conservation at an individual vertex. Susskind's answer is that such a question is misleading at this level of description. The full evolution conserves energy, but the interaction terms and the superposition of many

contributions are hidden when the process is compressed into a single diagram. If a diagram appears to violate naive on-shell energy conservation at a vertex, that is a sign that the picture is only bookkeeping, not a mechanical model.

He makes the same point again with the schematic electron-photon process. A local interaction can make an electron emit a photon and later reabsorb it. In perturbation theory this means that a physical electron has some amplitude to look like an electron plus a nearby photon. That is best interpreted as a correction to the structure of the electron, not as a dependable spacetime story of a tiny classical object literally spitting out and swallowing a photon. The more one insists on a mechanical picture, the more confused one becomes. The right way back is always the same: return to the expansion of $e^{-i\epsilon H}$ in powers of the Hamiltonian.

9.6 Restarting with Relativity

Only after all of this does Susskind return to the promise he made at the start of the course segment: fermions. Even here he does not begin with fermion fields. He begins with the Dirac equation itself. This marks a genuine change of subject: the lecture is now entering relativity.

The motivating example is the electron. Electrons are light, and in atoms their velocities are not always small compared to the speed of light. In hydrogen the effect is already noticeable; in heavy atoms it becomes substantially more important. So a relativistic equation for the electron is not optional if one wants quantitatively accurate atomic physics.

Susskind starts by recalling the usual nonrelativistic route. For a one-particle wavefunction, which we now denote by $\varphi(x,t)$ to avoid confusion with the earlier field operators,

$$E = \frac{p^2}{2m}. \tag{9.25}$$

Quantization means replacing

$$E \to i\partial_t, \qquad p \to -i\partial_x, \tag{9.26}$$

so that the Schrödinger equation becomes

$$i\partial_t \varphi = -\frac{1}{2m}\partial_x^2 \varphi. \tag{9.27}$$

For a relativistic particle, however,

$$E^2 = p^2 + m^2, \tag{9.28}$$

with $c = 1$. If one quantizes this relation directly, one obtains

$$-\partial_t^2 \varphi = \left(-\partial_x^2 + m^2\right)\varphi, \tag{9.29}$$

the Klein–Gordon equation. Susskind explicitly says that one might ask why Dirac did not simply stop there. Dirac's answer, in Susskind's retelling, is that quantum mechanics is supposed to be written in the form $i\partial_t \varphi = H\varphi$. He wants the Hamiltonian itself, not just its square.

Of course one could formally write

$$i\partial_t \varphi = \sqrt{-\partial_x^2 + m^2}\,\varphi, \tag{9.30}$$

$$H = \alpha P$$

$$i\,\frac{\partial}{\partial t}\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix} = -i\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix}\psi_1\\ \psi_2\end{pmatrix}$$

Figure 9.2: The two-component massless equation, matching the central board layout of the lecture.

but this introduces the square root of a differential operator, which Susskind describes as ugly and Dirac evidently found unacceptable as a fundamental starting point.

So Dirac restarts from the simplest possible relativistic case: a one-dimensional massless particle moving to the right. For such a particle,

$$E = P. \tag{9.31}$$

Quantization gives

$$i\partial_t\varphi = -i\partial_x\varphi, \tag{9.32}$$

or equivalently,

$$(\partial_t + \partial_x)\varphi = 0. \tag{9.33}$$

The solutions are

$$\varphi(x,t) = f(x-t), \tag{9.34}$$

so the wave profile moves rigidly to the right at the speed of light. Susskind likes this equation but immediately lists what is wrong with it. First, it allows negative energies when the momentum is negative. Second, it describes only right-moving particles. An ordinary electron cannot be a one-way street.

9.7 The One-Dimensional Dirac Equation

Dirac's first repair addresses only the left-right problem. Introduce two components,

$$\Psi(x,t) = \begin{pmatrix}\psi_1(x,t)\\ \psi_2(x,t)\end{pmatrix}, \tag{9.35}$$

where ψ_1 is the amplitude for the right-moving component and ψ_2 for the left-moving one. The operator that distinguishes the two is

$$\alpha = \begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}. \tag{9.36}$$

This is the Pauli matrix σ_3, but Susskind follows Dirac and calls it α. It is playing the role of a two-valued internal degree of freedom, analogous in form to spin but not literally spin.

The massless Hamiltonian is then

$$H = \alpha P, \tag{9.37}$$

and the equation of motion is

$$i\partial_t\Psi = -i\alpha\,\partial_x\Psi. \tag{9.38}$$

Written component by component, this says

$$i\partial_t \psi_1 = -i\partial_x \psi_1, \tag{9.39}$$
$$i\partial_t \psi_2 = +i\partial_x \psi_2. \tag{9.40}$$

So ψ_1 moves right and ψ_2 moves left. The one-way street has been cured. But Susskind immediately reminds the audience that this has *not* cured the negative-energy problem, and it has not yet given the electron a mass. In fact there is now an additional issue: both components are still massless, so both move at the speed of light.

Dirac's next step is the decisive one. Add a term proportional to the mass, but allow it to be a matrix acting on the two-component space:

$$H = \alpha p + \beta m. \tag{9.41}$$

The matrix β is not yet known. The requirement is that this Hamiltonian reproduce the relativistic energy-momentum relation when squared:

$$H^2 = p^2 + m^2. \tag{9.42}$$

Expanding,

$$\begin{aligned} H^2 &= (\alpha p + \beta m)^2 \\ &= \alpha^2 p^2 + \beta^2 m^2 + (\alpha\beta + \beta\alpha)\, pm. \end{aligned} \tag{9.43}$$

Here p commutes with α and β, since the momentum operator acts on the spatial dependence while the matrices act on the two-component index. Therefore the Dirac algebra is forced:

$$\alpha^2 = 1, \qquad \beta^2 = 1, \qquad \alpha\beta + \beta\alpha = 0. \tag{9.44}$$

These are exactly the conditions emphasized on the board.

Since α is already fixed, one may choose

$$\beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{9.45}$$

which indeed squares to the identity and anticommutes with α. The one-dimensional Dirac equation is then

$$i\partial_t \Psi = (-i\alpha\partial_x + \beta m)\, \Psi. \tag{9.46}$$

This is the form Susskind identifies at the end of the lecture as the Dirac equation in one dimension. He remarks that a more symmetric and more standard-looking form is obtained by multiplying through by β, but postpones that rewriting until the next lecture.

The component form is especially instructive:

$$i\partial_t \psi_1 = -i\partial_x \psi_1 + m\psi_2, \tag{9.47}$$
$$i\partial_t \psi_2 = +i\partial_x \psi_2 + m\psi_1. \tag{9.48}$$

The reason these equations are new is that the mass term is off-diagonal: it literally swaps the two components. In the language raised by a student near the end of the lecture, the chirality is

measured by α, while the mass term is the piece that mixes the two chiralities. Susskind explicitly agrees that the mass term is the chiral oscillation.

He then adds an important physical comment. A truly massless one-dimensional particle moving to the right stays right-moving under Lorentz transformations; in that sense the right-moving and left-moving sectors can remain separate. A massive particle is different. Because one can always boost to a frame in which its velocity changes sign, a purely right-moving massive particle is impossible. The mass issue and the left-right issue are therefore not independent. In this simple one-dimensional construction, fixing the latter naturally provides the mechanism for the former.

Still, the lecture ends with clear unfinished business. The negative-energy problem remains; in Susskind's own phrase, the construction has in that sense made things worse, because now there are negative-energy right-movers and negative-energy left-movers. And the theory is still only one-dimensional. The move to $3+1$ dimensions, and the more familiar covariant form of the Dirac equation, are deferred.

9.8 Summary

The lecture begins by revisiting the second-quantized Schrödinger Hamiltonian and using it, very explicitly, to show how momentum conservation comes from the integral over space. The Fourier transform turns locality into a delta function, and the delta function enforces conservation of total momentum. Once that mechanism is clear, local products of fields become natural candidates for interaction terms suggested by experiment. The examples of contact scattering and decay then lead directly to the perturbative expansion of time evolution, where H^2, H^3, and higher powers generate exchange processes, self-energy corrections, and the first diagrammatic language of quantum field theory.

Only after that does Susskind restart from relativity and build the one-dimensional Dirac equation. The path is deliberate: reject the ugly square-root Hamiltonian, begin with a massless right-mover, cure the one-way street by introducing a two-component wavefunction and the matrix α, then add a mass term βm and demand that squaring the Hamiltonian reproduce $E^2 = p^2 + m^2$. The result is the one-dimensional Dirac equation, with mass appearing as the coupling between left- and right-moving sectors. What the lecture does *not* yet solve is equally important: negative energies remain, and the full three-dimensional theory is still to come.

Chapter 10

Fermions, Dirac Matrices, and the Dirac Sea

This lecture begins with a promise and an interruption. Leonard Susskind opens by invoking the Dirac sea, but before going there he deliberately backs up: first to the difference between bosons and fermions in second quantization, then to the simpler one-dimensional Dirac construction, and only then to the three-dimensional equation appropriate to the physical world. That order matters. The later discussion of holes and positrons depends on the earlier algebra of fermions, and the later 4×4 Dirac matrices are introduced only after the 2×2 theory has been pushed until it fails. These notes follow that progression closely. Throughout, we normalize the lecture's shifting notation by writing operator equations with P or $\mathbf{P}$, and eigenvalue relations with p or $\mathbf{p}$. We use units $c = \hbar = 1$.

10.1 Fermionic Exchange and Anti-Commutation

Before returning to the Dirac equation, Susskind asks how many-particle systems are built from field operators, and in particular how fermions differ from bosons. The first point is familiar but important: bosons may occupy the same state, while fermions may not. He wants that statement to arise from operator algebra.

Take a two-particle state created from the vacuum,

$$|x, y\rangle = \psi^\dagger(x)\psi^\dagger(y)|0\rangle. \tag{10.1}$$

Here x and y label the positions of particle 1 and particle 2; they are not the x- and y-coordinates of one particle. For bosons, exchanging the two particles changes nothing:

$$|x, y\rangle = |y, x\rangle. \tag{10.2}$$

At the operator level this is implemented by commuting creation operators,

$$\psi^\dagger(x)\psi^\dagger(y) = \psi^\dagger(y)\psi^\dagger(x). \tag{10.3}$$

For fermions the exchanged state picks up a minus sign,

$$|x, y\rangle = -|y, x\rangle. \tag{10.4}$$

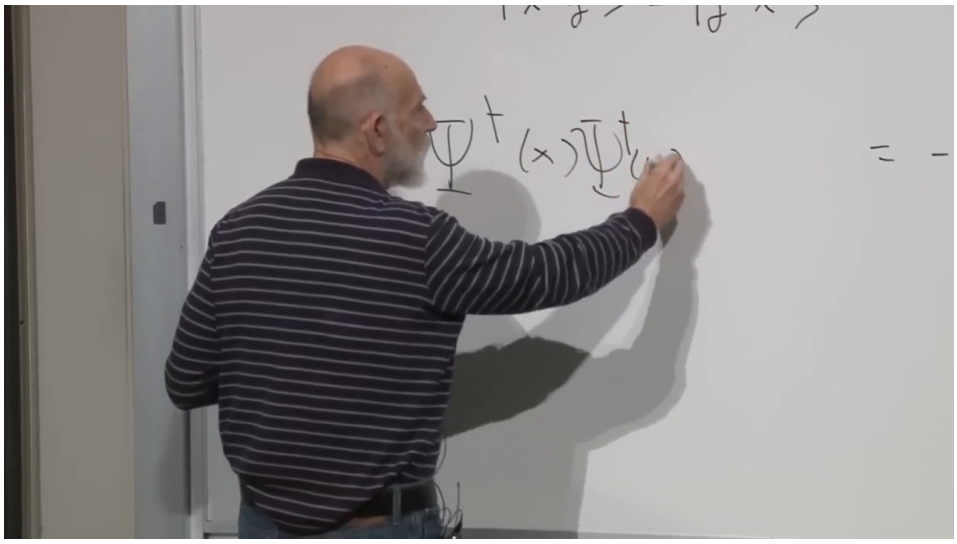


Figure 10.1: Fermionic minus-sign operator step from exchange antisymmetry. The right-hand side is partly hidden on the board, so the notes use the transcript-backed completion $\psi^\dagger(x)\psi^\dagger(y) = -\psi^\dagger(y)\psi^\dagger(x)$.

An overall sign of a state vector is not itself observable, but the sign is crucial in the formulas, and Susskind's point is that the formulas force the field operators to anti-commute. The transcript-backed completion of the partially obscured board equation is

$$\psi^\dagger(x)\psi^\dagger(y) = -\psi^\dagger(y)\psi^\dagger(x). \tag{10.5}$$

Definition 10.1. For two operators A and B, the anti-commutator is

$$\{A, B\} = AB + BA. \tag{10.6}$$

So the fermionic relation is equivalently

$$\{\psi^\dagger(x), \psi^\dagger(y)\} = 0. \tag{10.7}$$

This is the lecture's first decisive step: exchange antisymmetry is translated into the algebra of creation operators.

Susskind then immediately specializes to the case $y = x$. This is the lecture's route to the exclusion principle. Starting from the anti-commutator,

$$\{\psi^\dagger(x), \psi^\dagger(x)\} = 0 \Longrightarrow \psi^\dagger(x)\psi^\dagger(x) + \psi^\dagger(x)\psi^\dagger(x) = 0 \tag{10.8}$$

$$\Longrightarrow 2\,\psi^\dagger(x)\psi^\dagger(x) = 0 \tag{10.9}$$

$$\Longrightarrow (\psi^\dagger(x))^2 = 0. \tag{10.10}$$

Trying to create two identical fermions in the same state gives the zero vector, not a new state.

Because the board is visually important here, it is worth pairing it with a clean reconstruction of the logical chain:

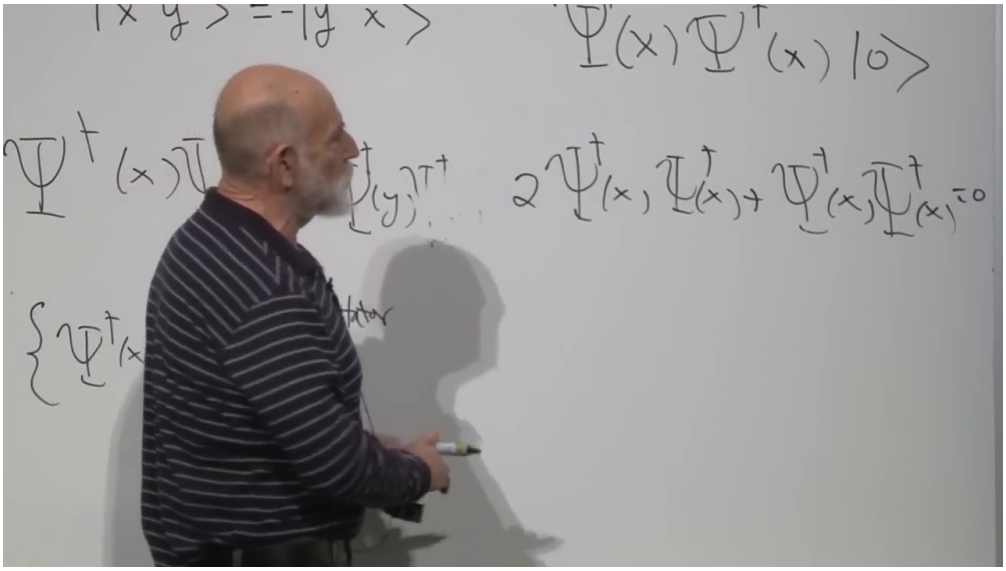


Figure 10.2: Equal-point anti-commutator and exclusion-principle simplification. The visible board equation already shows the factor 2, the repeated creation-operator product at x, and the vanishing result.

$$\{\psi^\dagger(x), \psi^\dagger(x)\} = 0 \longrightarrow 2(\psi^\dagger(x))^2 = 0 \longrightarrow (\psi^\dagger(x))^2 = 0$$

Figure 10.3: Clean reconstruction of the equal-point fermionic exclusion step.

10.2 Occupation Numbers and the Filled/Empty Symmetry

Susskind does not move on as soon as exclusion has been obtained. He lingers on the algebra long enough to extract a second idea, one that he explicitly says will matter later: for a single fermionic mode, creation and annihilation operators are algebraically more symmetric than for bosons.

For a single bosonic mode one may summarize the usual structure by

$$[a, a^\dagger] = 1, \qquad n_{\text{boson}} = 0, 1, 2, \ldots \tag{10.11}$$

The occupation-number ladder stretches upward without bound. The annihilation operator eventually reaches the bottom and gives zero; the creation operator never runs out of states above it. In that very concrete sense the bosonic ladder distinguishes "up" from "down."

For a single fermionic mode the algebra is instead

$$\{a, a^\dagger\} = 1, \qquad a^2 = (a^\dagger)^2 = 0, \qquad n_{\text{fermion}} = 0, 1. \tag{10.12}$$

There are only two possibilities: empty or filled. The creation operator takes 0 to 1, but acting once more gives zero because there is no state above 1. The annihilation operator takes 1 to 0, but acting once more gives zero because there is no state below 0.

So the single-mode fermionic spectrum is symmetric in a way the bosonic spectrum is not. Empty and filled exchange roles under $a \leftrightarrow a^\dagger$. Physically the two operators still do different things, but algebraically one cannot tell them apart as sharply as in the bosonic case. Susskind pauses here to emphasize that this is not yet a claim about particles and anti-particles; it is simply a structural

fact about fermionic operators. But it is exactly the fact that later makes the hole interpretation feel natural rather than miraculous.

Remark 10.2. In the lecture Susskind comes back to this point much later, when the annihilation of a negative-energy electron is reinterpreted as the creation of a positron. The symmetry between filled and empty states is prepared here, not inserted retroactively at the end.

10.3 Review of the One-Dimensional Dirac Construction

Having put the fermionic operator algebra in place, Susskind says, in effect, let us go back to the Dirac equation. For the moment he again works at the single-particle level.

The starting point is the one-dimensional massless equation

$$i\partial_t\psi = -\,i\partial_x\psi. \tag{10.13}$$

This is a Schrödinger equation with Hamiltonian

$$H = P, \tag{10.14}$$

so on momentum eigenstates

$$E = p. \tag{10.15}$$

Wave packets move rigidly to the right. It is a perfectly good relativistic equation for a massless right-mover, but it is not yet a satisfactory model of a massive electron.

The relativistic target is

$$E^2 = p^2 + m^2. \tag{10.16}$$

Before adding mass, however, Susskind first doubles the one-dimensional theory by introducing a left-moving partner:

$$H\psi_1 = P\psi_1, \qquad H\psi_2 = -P\psi_2. \tag{10.17}$$

Now one has two species, one always moving to the right and one always moving to the left. Packaging them into a two-component wavefunction,

$$\psi = \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix}, \tag{10.18}$$

one introduces a matrix α with eigenvalues $+1$ and -1,

$$\alpha = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{10.19}$$

so that both equations are summarized by

$$H\psi = \alpha P\,\psi. \tag{10.20}$$

This doubles the number of components and introduces an internal two-valued degree of freedom. In the one-dimensional lecture review Susskind interprets it as handedness: right-moving versus left-moving.

Mass still has not appeared. To generate it, he introduces a second matrix β which couples the two sectors. In the one-dimensional construction the convenient choice is

$$\beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \tag{10.21}$$

and one writes

$$H = \alpha P + \beta m. \tag{10.22}$$

The required algebra is

$$\alpha^2 = 1, \qquad \beta^2 = 1, \qquad \alpha\beta + \beta\alpha = 0. \tag{10.23}$$

Susskind explicitly warns that this is matrix anti-commutation, not the anti-commutation of fermionic fields.

Now the worked derivation is straightforward:

$$\begin{aligned} H^2 &= (\alpha P + \beta m)^2 && (10.24)\\ &= \alpha^2 P^2 + \alpha\beta P m + \beta\alpha m P + \beta^2 m^2 && (10.25)\\ &= \alpha^2 P^2 + (\alpha\beta + \beta\alpha) P m + \beta^2 m^2 && (10.26)\\ &= P^2 + m^2. && (10.27) \end{aligned}$$

Thus the mass term works because the cross term cancels. Susskind's physical gloss is that the off-diagonal matrix couples left-moving and right-moving sectors. He immediately qualifies the picture, but the picture is still useful: mass behaves like a term that mixes the two handed motions, and that is why a massive particle can be thought of as capable, in some averaged sense, of being at rest.

10.4 From One Dimension to Three: $H = \boldsymbol{\alpha} \cdot \mathbf{P}$

The lecture's next hard pivot is explicit: how do we do this in three dimensions?

The one-dimensional Hamiltonian $H = \alpha P$ cannot simply be copied, because now momentum is a three-vector. The board writes

$$H = \alpha \cdot P, \tag{10.28}$$

and Susskind's point is that α must therefore become a vector of matrices. In normalized notation we write the operator equation as

$$H = \boldsymbol{\alpha} \cdot \mathbf{P} = \alpha_x P_x + \alpha_y P_y + \alpha_z P_z. \tag{10.29}$$

For a massless relativistic particle the desired dispersion relation is

$$E^2 = \mathbf{p}^2 = p_x^2 + p_y^2 + p_z^2. \tag{10.30}$$

Accordingly one wants the operator identity

$$H^2 = \mathbf{P}^2. \tag{10.31}$$

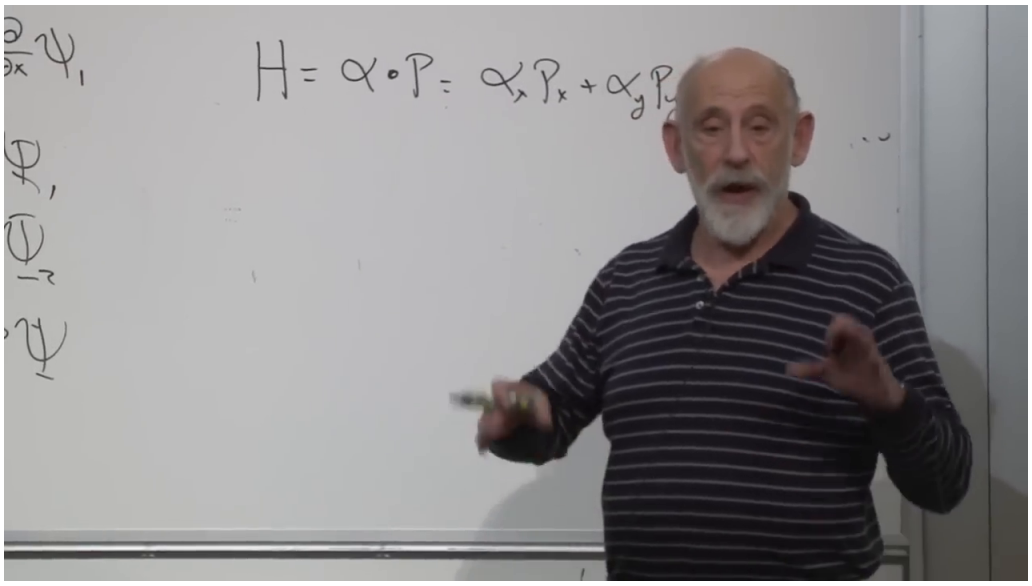


Figure 10.4: Three-dimensional Dirac ansatz $H = \boldsymbol{\alpha} \cdot \mathbf{p}$. The board writes $H = \alpha \cdot P$; the notes normalize this to the operator form $H = \boldsymbol{\alpha} \cdot \mathbf{P}$.

Squaring the Hamiltonian gives

$$H^2 = \alpha_x^2 P_x^2 + \alpha_y^2 P_y^2 + \alpha_z^2 P_z^2 \tag{10.32}$$
$$+ (\alpha_x \alpha_y + \alpha_y \alpha_x) P_x P_y \tag{10.33}$$
$$+ (\alpha_x \alpha_z + \alpha_z \alpha_x) P_x P_z \tag{10.34}$$
$$+ (\alpha_y \alpha_z + \alpha_z \alpha_y) P_y P_z. \tag{10.35}$$

Thus the matrices must satisfy

$$\alpha_x^2 = \alpha_y^2 = \alpha_z^2 = 1, \qquad \alpha_i \alpha_j + \alpha_j \alpha_i = 0 \quad (i \neq j). \tag{10.36}$$

The cross terms vanish only if the different α_i mutually anti-commute.

Here the lecture arrives at the Pauli matrices. They satisfy exactly the required algebra:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{10.37}$$

So one may take

$$\alpha_x = \sigma_x, \qquad \alpha_y = \sigma_y, \qquad \alpha_z = \sigma_z. \tag{10.38}$$

The Hamiltonian then becomes

$$H = \boldsymbol{\sigma} \cdot \mathbf{P}. \tag{10.39}$$

A clean way to state the resulting spectrum is

$$\boldsymbol{\sigma} \cdot \mathbf{p} = |\mathbf{p}| \, (\boldsymbol{\sigma} \cdot \hat{\mathbf{p}}), \qquad \hat{\mathbf{p}} = \frac{\mathbf{p}}{|\mathbf{p}|}. \tag{10.40}$$

Since $\boldsymbol{\sigma} \cdot \hat{\mathbf{p}}$ has eigenvalues ± 1, the energy eigenvalues are

$$E = \pm |\mathbf{p}|. \tag{10.41}$$

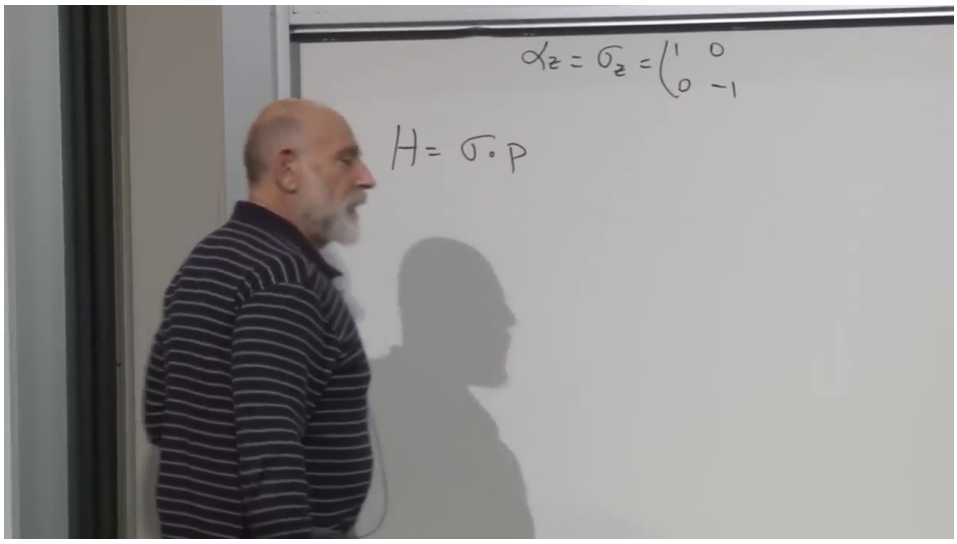


Figure 10.5: Massless Hamiltonian summarized as $H = \boldsymbol{\sigma} \cdot \mathbf{p}$. The board's compact formula follows the explicit component-by-component identification of the three α_i with the Pauli matrices.

This is the mathematically clean form of Susskind's explanation that the energy is the component of spin along the momentum direction. Positive energy corresponds to spin aligned with momentum; negative energy corresponds to spin anti-aligned with momentum.

Remark 10.3. The transcript is corrupted at the point where Susskind responds to the claim that spin somehow "drops out" of the Dirac equation. The surrounding logic is nevertheless clear. Dirac wanted a Hamiltonian linear in momentum; ordinary numbers could not do the job; the Pauli matrices were forced on him by the requirement $H^2 = \mathbf{P}^2$. In that sense spin is not tacked on afterward but emerges from the linearization logic itself.

10.5 Why Mass Forces 4×4 Matrices

At this stage Susskind immediately asks the next question: if the 2×2 massless theory works so neatly, why not just add a mass term to it? The natural trial is

$$H = \boldsymbol{\sigma} \cdot \mathbf{P} + \beta m. \tag{10.42}$$

Assuming again that $\beta^2 = 1$, squaring H would give the desired m^2 term, but only if the cross term vanishes. That requires

$$\sigma_i \beta + \beta \sigma_i = 0 \qquad (i = x, y, z). \tag{10.43}$$

So β would have to anti-commute with all three Pauli matrices.

That is precisely what fails. In 2×2 matrices the Pauli matrices already exhaust the family of mutually anti-commuting matrices. There is no further 2×2 matrix that anti-commutes with all three. This is the lecture's algebraic obstruction: the three-dimensional 2×2 equation is intrinsically massless.

The next step is therefore not a matter of taste. One must enlarge the internal vector space and look for 4×4 matrices. Susskind emphasizes that the first matrix size supporting four mutually

anti-commuting matrices is 4×4, and that this is a mathematical fact about matrix algebras, not a consequence of spacetime having four dimensions.

A convenient representation, consistent with the block description given in the lecture, is

$$\alpha_i = \begin{pmatrix} \sigma_i & 0 \\ 0 & -\sigma_i \end{pmatrix}, \qquad \beta = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}. \tag{10.44}$$

These satisfy

$$\{\alpha_i, \alpha_j\} = 2\delta_{ij}, \qquad \{\alpha_i, \beta\} = 0, \qquad \beta^2 = 1. \tag{10.45}$$

The Hamiltonian becomes

$$H = \boldsymbol{\alpha} \cdot \mathbf{P} + \beta m, \tag{10.46}$$

or explicitly

$$H = \begin{pmatrix} \boldsymbol{\sigma} \cdot \mathbf{P} & mI \\ mI & -\boldsymbol{\sigma} \cdot \mathbf{P} \end{pmatrix}. \tag{10.47}$$

Its square is then

$$H^2 = \mathbf{P}^2 + m^2. \tag{10.48}$$

Remark 10.4. The lecture moves quickly through the explicit matrices and repeatedly reminds the audience that many equivalent representations exist. The representation above should therefore be read as a cautious standard reconstruction matching Susskind's block-wise description, not as a uniquely privileged choice.

What matters is the structure: the massive three-dimensional Dirac equation needs a four-component spinor because the 2×2 matrix algebra is too small to accommodate the extra anti-commuting matrix required by the mass term.

10.6 Chirality, Velocity, and Spin

Once the 4×4 theory is on the board, the lecture turns to interpretation. Several student questions now become part of the mathematical story.

Chirality and the mass term. In the massless 2×2 theory, positive energy ties the spin to the momentum. In the lecture Susskind speaks of chirality as the relation between the two, so that one alignment is right-handed and the opposite alignment is left-handed. In more standard language, for the massless case this is the same idea that is often called helicity.

The crucial point is that the 2×2 massless theory contains only one handedness at a time. The 4×4 enlargement brings in the opposite handedness as well, and the off-diagonal βm term mixes the two blocks. That is why Susskind repeatedly describes the mass term as a coupling between left-handed and right-handed sectors. For a massless particle the handedness is conserved; with the mass term present, transitions between the two become possible.

Velocity from the Heisenberg equation. Susskind then asks a more surprising question: what physical observable is represented by the matrices α_i? His answer is that α_i are the components of the velocity operator.

Using the Heisenberg equation in the standard form

$$\dot{L} = i[H, L], \tag{10.49}$$

together with the canonical commutator

$$[x, P_x] = i, \tag{10.50}$$

we compute

$$\begin{aligned} \dot{x} &= i[H, x] && (10.51)\\ &= i[\alpha_x P_x + \alpha_y P_y + \alpha_z P_z + \beta m,\ x] && (10.52)\\ &= i\alpha_x [P_x, x] && (10.53)\\ &= \alpha_x. && (10.54) \end{aligned}$$

The matrices commute with x because they act in the internal spinor space, not in position space. So only P_x contributes. The conclusion is the lecture's punchline:

$$\dot{x} = \alpha_x. \tag{10.55}$$

Remark 10.5. Susskind explicitly says that he does not remember the sign convention in the Heisenberg equation and does not care to stop and fix it. The convention used here is the standard one. The only indispensable lecture claim is the resulting identification $\dot{x} = \alpha_x$.

Each α_i has eigenvalues ± 1, so an instantaneous measurement of a velocity component yields ± 1 in our units. But that is not the end of the story, because the velocity operator is not conserved:

$$\dot{\alpha}_x = i[H, \alpha_x] \neq 0. \tag{10.56}$$

The Hamiltonian contains α_y and α_z, and they do not commute with α_x. Hence the velocity oscillates. This is Susskind's discussion of *zitterbewegung*: the rapid trembling motion of the free Dirac electron.

He is careful to separate this from the conservation of momentum. Newton's law says that the momentum of a free particle is constant; it does not say that every operator one might call "velocity" must be constant. In this theory momentum and velocity are not the same operator. The lecture keeps the explanation qualitative, but the basic point is that the time average of the rapidly fluctuating velocity reproduces the expected smooth motion, and for slow particles that average velocity is P/m.

Velocity is not spin. A further student question prompts an important clarification. The matrices α_i are not the genuine spin operators. They transform as an ordinary vector; spin is an axial vector. In the block notation above, a natural corresponding spin operator is

$$\Sigma_i = \begin{pmatrix} \sigma_i & 0 \\ 0 & \sigma_i \end{pmatrix}, \tag{10.57}$$

which differs from α_i by the sign in the lower block. Susskind's point is that the Dirac electron carries more internal structure than a single naive "spin" label. One binary distinction is tied to handedness; another is the actual spin of the electron.

The lecture briefly detours through quaternions because a student asks about them. The only lasting point for us is that the Pauli matrices sit inside a very rigid algebraic structure. The quaternion aside itself is not part of the main mathematical spine, and Susskind does not use it in what follows.

10.7 Negative Energy, the Dirac Sea, and Positrons

Only now does the lecture finally return to its announced destination. Susskind immediately simplifies the setting again. To discuss negative energy one does not need the full $3+1$-dimensional Dirac equation; the essential issue is already present in the simplest one-dimensional right-moving massless case,

$$H = P. \tag{10.58}$$

Because P can take either sign, the theory contains both positive- and negative-energy states.

This is dangerous. The vacuum should be the state of lowest energy. If one can keep adding negative-energy particles, then the energy can be driven downward without bound and there is no stable vacuum. Susskind's resolution is historical and explicitly Diracian: since the particles are fermions, every negative-energy state can be occupied at most once. Therefore the lowest-energy state is obtained by filling all negative-energy states. That filled configuration is the Dirac sea.

The lecture stresses that if the particles were bosons this would fail. One could put arbitrarily many bosons into the same negative-energy state and the energy would continue to decrease. So the stability of the Dirac vacuum depends crucially on Fermi statistics.

Once the negative-energy states are filled, two kinds of excitations remain. One may add an ordinary positive-energy electron above the sea. Or one may remove a negative-energy electron from the sea. The removed electron leaves a hole. Since the missing object carried negative charge, the hole carries positive charge. Since removing a negative-energy particle raises the energy, the hole has positive energy:

$$E_{\text{hole}} = -E_{\text{negative electron}} > 0. \tag{10.59}$$

That hole is the positron.

At this point the lecture returns to the earlier algebraic symmetry of fermionic operators. The field operator is first written schematically as an integral over all momenta,

$$\psi(x) \sim \int_{-\infty}^{\infty} dp\, a(p) e^{-ipx}. \tag{10.60}$$

Susskind then splits the integral into positive- and negative-momentum pieces,

$$\psi(x) \sim \int_{0}^{\infty} dp\, a(p) e^{-ipx} + \int_{-\infty}^{0} dp\, a(p) e^{-ipx}. \tag{10.61}$$

The second piece annihilates negative-energy electrons. Using the fermionic symmetry between filled and empty states, one reinterprets that operator as a positron creation operator. Writing schematically

$$b^\dagger(p) \equiv a(-p), \tag{10.62}$$

the field becomes

$$\psi(x) \sim \int_{0}^{\infty} dp\, a(p) e^{-ipx} + \int_{0}^{\infty} dp\, b^\dagger(p) e^{ipx}. \tag{10.63}$$

The first term annihilates positive-energy electrons; the second creates positrons. The precise normalization and phase conventions are not the point here. The lecture's point is the structural split.

Susskind then inserts this idea into the schematic interaction term

$$H_{\text{int}} \sim \psi^\dagger(x)\psi(x)A(x). \tag{10.64}$$

Because ψ and $\psi^\dagger$ each contain both electron and positron pieces, the same operator contains several processes at once. Depending on which pieces are selected, it can describe

1. an electron emitting or absorbing a photon,
2. an electron and a positron annihilating into a photon,
3. a photon creating an electron-positron pair.

This is Susskind's way of showing that a single field-theoretic coupling encodes many different channels, which later become the lines and vertices of Feynman-diagram language.

The lecture then shifts from Dirac's historical picture to the more modern reinterpretation. Modern physicists usually do not describe the vacuum as literally full of negative-energy electrons. Instead they simply flip the operator assignments in the negative-energy sector from the start. Creation operators for negative-energy electrons are replaced by annihilation operators for positive-energy positrons, and the symmetry between electrons and positrons becomes manifest at once. Susskind nevertheless keeps Dirac's sea in the foreground because it captures the logic of the discovery.

He closes with the condensed-matter analogy. In a crystal, electrons fill states up to the Fermi level; removing one leaves a hole; that hole behaves as a positive charge carrier with positive energy measured relative to the filled ground state. The pattern is strikingly similar: a filled sea, particles above it, and holes within it. The analogy is pedagogically valuable, even though historically Dirac arrived at his picture before the solid-state language became familiar.

Finally, in response to questions, Susskind distinguishes the fermionic Dirac story from the bosonic Klein–Gordon story. The Klein–Gordon equation is structurally different; it is not simply a Hamiltonian linear in momentum with Dirac matrices, and he does not attempt to fold it into the same narrative in this lecture.

10.8 Summary

The lecture unfolds as a chain of necessities rather than a list of topics. Fermionic exchange symmetry leads to anti-commuting creation operators, and the equal-point specialization gives exclusion. The two-state fermionic occupation number then prepares the later particle-hole reinterpretation. Only after that groundwork is in place does Susskind review the one-dimensional Dirac equation, add the left-moving partner, and show that a mass term is a coupling that mixes the two sectors.

From there the move to three dimensions is forced by rotational invariance. The demand that a Hamiltonian be linear in momentum and yet satisfy $H^2 = \mathbf{P}^2$ produces the Pauli matrices and the massless Hamiltonian $H = \boldsymbol{\sigma} \cdot \mathbf{P}$. The failure of any 2×2 mass term then forces the jump to 4×4 Dirac matrices, where mass becomes a coupling between opposite chiralities. The interpretation of α as velocity, the nonconservation of velocity, and the distinction between α and true spin are all consequences of that enlarged structure.

Only after those algebraic and kinematical steps are laid down does the original promise return. Negative energy is stabilized by Fermi statistics, the filled Dirac sea becomes the vacuum, holes become positrons, the fermion field splits into electron-annihilation and positron-creation parts, and a simple interaction term already contains emission, annihilation, and pair creation. The lecture thus ends where it began, but with all the necessary machinery now in place.